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RESEARCH ARTICLE

Probabilistic Multivariate Statistical Process Control via Kernel Parameter Uncertainty Propagation

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ABSTRACT Kernel-based multivariate statistical process control (K-MSPC) extends classical latent-variable monitoring to nonlinear industrial processes, but its performance depends strongly on kernel parameters such as lengthscales and variance terms. In current practice, these parameters are typically selected using heuristic rules or deterministic optimisation and are then treated as fixed, despite being estimated from finite and noisy data. This may lead to overconfident control limits and unstable alarm behaviour when the kernel configuration is uncertain. This work proposes a probabilistic K-MSPC framework for quantifying and propagating kernel parameter uncertainty to process monitoring statistics and diagnostic contributions. The proposed method follows a two-stage workflow: first, kernel parameters are calibrated deterministically using supervised or unsupervised models; second, Bayesian inference is performed using Markov chain Monte Carlo to obtain posterior samples of the kernel parameters. These posterior samples are propagated through kernel principal component analysis to generate probabilistic Hotelling's T^2 and squared prediction error control charts, together with uncertainty-aware contribution plots. The framework is evaluated using the Tennessee Eastman Process benchmark. The results show that posterior-mean monitoring improves fault detection compared with deterministic prior-mean charts for the squared exponential (SE) kernel in several fault scenarios. The bands remain narrow during normal operation and widen under faulty conditions, indicating that epistemic uncertainty is amplified in abnormal regimes. The automatic relevance determination (ARD) kernel performs better than the single-length SE kernel, reduces posterior uncertainty, whereas the unsupervised calibration route produces wider posterior bands while maintaining robust fault detection.

INDEX TERMS Kernel PCA, Bayesian inference, MCMC, uncertainty quantification, Tennessee Eastman process.

I. INTRODUCTION

Modern industrial plants are increasingly instrumented with high-frequency, high-dimensional measurements, creating a strong need for reliable data-driven monitoring methods that can detect abnormal operating conditions early and support diagnosis [1]. More broadly, uncertainty-aware monitoring is closely connected to the development of reliable fault-tolerant and robust control systems, where fault detection, fault estimation, and robust decision-making are essential for maintaining safe operation under abnormal conditions [2], [3], [4].

Multivariate statistical process control (MSPC) remains a cornerstone in this space, where latent-variable models such as principal component analysis (PCA) translate multivariate measurements into low-dimensional monitoring statistics and

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corresponding control charts, typically Hotelling's T^2 and squared prediction error (SPE) indicators, for fault detection and interpretation [5], [6]. In contemporary manufacturing settings, MSPC is frequently embedded within broader ''data science for operations'' and Industry 4.0 initiatives, where statistical process control and related monitoring practices are combined with modern analytics workflows [7].

While classical MSPC based on PCA has proven effective for many industrial applications, its linear structure limits its ability to capture complex nonlinear relationships commonly encountered in modern processes [8]. To address this limitation, nonlinear extensions such as Kernel PCA (K-PCA) are introduced by mapping data into a high-dimensional reproducing kernel Hilbert space (RKHS), where linear projections correspond to nonlinear structures in the original space [8]. Kernel-based MSPC (K-MSPC) has since demonstrated improved fault detection capabilities in processes exhibiting nonlinear dynamics, multimodality, and complex interactions [9], [10], [11].

However, this increased modelling flexibility introduces an additional layer of complexity: the selection of kernel parameters [49], such as the length scale and variance parameters in radial basis function kernels. These parameters directly influence the geometry of the feature space, the variance captured by principal components, and ultimately the sensitivity of T^2 and SPE monitoring statistics [12]. In most existing K-MSPC studies, kernel parameters are selected using heuristic rules (e.g., median distance criteria), cross-validation, or deterministic optimisation procedures [13], [14]. Once chosen, they are treated as fixed quantities, and subsequent monitoring proceeds under the implicit assumption that the selected kernel represents the true data-generating structure.

Optimisation procedures such as gradient-based methods, genetic algorithms, or heuristic selection rules yield point estimates that are subsequently treated as fixed and certain [15]. However, kernel parameters are not directly observable quantities; they are inferred from finite and often noisy training data. As such, they are subject to epistemic uncertainty arising from limited sample size, model misspecification, and optimisation landscape nonconvexity.

Ignoring this uncertainty has important implications for statistical process monitoring. The geometry of the kernel feature space — and therefore the eigenstructure of the K-PCA model — depends sensitively on the chosen parameters. Small variations in the lengthscale, for instance, can significantly alter the variance distribution between the retained principal subspace and the residual space [14]. Consequently, control limits derived from deterministic kernel estimates may be overconfident, potentially underestimating variability under normal conditions or mischaracterising fault persistence. This can manifest as unstable alarm behaviour, inflated false alarms, or premature return-to-normal indications in control charts. In other words, while K-MSPC extends linear MSPC to nonlinear settings, it inherits a hidden layer of uncertainty that remains unquantified in standard practice.

To address this limitation, the objective of this work is to develop a probabilistic kernel-based multivariate statistical process control framework that explicitly quantifies and propagates uncertainty in kernel parameters to monitoring statistics. Rather than treating kernel parameters as fixed point estimates, we model them as random variables and infer their posterior distributions using Bayesian techniques.

The resulting uncertainty is then propagated through the Kernel PCA model to obtain probabilistic control charts for both the T^2 and SPE statistics, as well as for their corresponding variable contribution diagnostics. By embedding kernel calibration within a Bayesian inference framework, the proposed approach enables monitoring decisions that reflect epistemic uncertainty in model structure. This leads to control charts that not only signal faults but also quantify confidence in detection and attribution, thereby enhancing robustness, interpretability, and reliability in nonlinear process monitoring.

The proposed framework is also complementary to recent developments in robust and adaptive control. Advanced nonlinear control strategies, such as practical finite-time command-filtered adaptive backstepping and adaptive neural zeta-backstepping, have been developed to improve closed-loop performance under uncertainty, disturbances, and nonlinear dynamics [22], [23]. While the present work does not address controller synthesis, reliable uncertainty-aware monitoring can provide useful information for such downstream decision-making tasks by identifying abnormal operating regimes and quantifying confidence in fault detection and variable attribution.

A. OUR CONTRIBUTION

This work is a direct continuation of Duma et al. [14], which addressed deterministic kernel calibration for K-MSPC through supervised and procedural optimisation. The present paper addresses the complementary question left open there: how uncertainty in the calibrated kernel parameters propagates to monitoring statistics, control limits, and contribution diagnostics.

The main contribution of this paper is:

- Aiding process monitoring using control charts to be to accurate under uncertain and nonlinear conditions, thereby enhancing industrial quality yield and productivity.

This contribution is achieved by the following sub-contributions:

1) We propose a two-stage probabilistic calibration framework for K-MSPC. In the first stage, kernel parameters are estimated deterministically using both supervised and unsupervised strategies. The supervised route employs intermediary classification or regression models (Gaussian process classification, GPC [41], and Kernel PCR [14]) optimised via gradient-based and derivative-free methods, while the unsupervised route evaluates a suite of kernel-selection criteria

tailored to healthy-operation data. This stage provides informed prior means for subsequent Bayesian inference.
2) We introduce Bayesian sampling of kernel parameters using Markov chain Monte Carlo (MCMC) methods, including adaptive Metropolis (AM) [35], delayed rejection adaptive Metropolis (DRAM) [34], Hamiltonian Monte Carlo (HMC) [26], and the No-U-Turn Sampler (NUTS) [37]. This enables posterior inference over kernel lengthscales and variance parameters rather than reliance on single-point estimates.
3) We formulate probabilistic T^2 and SPE control charts by propagating posterior uncertainty through the Kernel PCA monitoring framework. In addition to probabilistic monitoring statistics, we derive uncertainty-aware contribution plots, allowing variable attribution to be expressed together with credible intervals.

The proposed framework differs conceptually from three related research directions. First, while Bayesian inference of kernel parameters is well established in Gaussian process modelling, existing work focuses on predictive uncertainty of the GP output rather than on how parameter uncertainty alters downstream monitoring structure. Here, the posterior distribution is not an end in itself but a mechanism for reshaping the K-PCA eigenspace and the associated monitoring statistics.

Second, unlike conventional K-MSPC approaches that rely on a single optimised kernel configuration, the proposed formulation replaces fixed-kernel monitoring with a distribution over monitoring models, thereby exposing the sensitivity of control charts to kernel calibration.

Third, in contrast to uncertainty bands derived from bootstrap resampling or measurement noise around a fixed model, the uncertainty quantified in this work is structural and epistemic: it originates from kernel calibration itself and propagates through both detection statistics and diagnostic contributions. This perspective reframes kernel selection not as a preprocessing step but as a source of model uncertainty that directly affects alarm stability, fault persistence, and interpretability.

The novelty of the proposed framework lies in the propagation of posterior kernel-parameter uncertainty through the complete K-MSPC monitoring pipeline. While Bayesian inference of kernel parameters is well established in Gaussian process modelling [50], its usual purpose is to quantify predictive uncertainty in a supervised input–output model. In contrast, the present work uses the posterior distribution of the kernel parameters to induce a distribution over K-PCA monitoring models. Each posterior draw changes the kernel matrix, the centered feature-space geometry, the K-PCA eigenspace, and consequently the resulting T^2 and SPE statistics, control-chart behaviour, and contribution diagnostics. Therefore, the uncertainty quantified here is not only output-prediction uncertainty, but uncertainty in the nonlinear monitoring structure itself.

This also distinguishes the proposed method from probabilistic PCA and bootstrap-based monitoring. Probabilistic PCA introduces a probabilistic formulation of a linear latent-variable model, whereas the present framework addresses uncertainty in the kernel-induced nonlinear feature-space geometry used by K-MSPC. Bootstrap monitoring methods [51] quantify variability due to resampling of observations, while the proposed approach targets epistemic uncertainty in the kernel parameters after calibration. As a result, the proposed framework provides uncertainty-aware control charts and contribution plots that directly express how sensitive fault detection and fault attribution are to kernel calibration uncertainty.

The framework is evaluated using the Tennessee Eastman process (TEP) [28] benchmark, demonstrating how parameter uncertainty affects fault detection performance, control-limit stability, and interpretability under both supervised and unsupervised calibration schemes.

B. OUTLINE OF THE PAPER

This paper is organized as follows: the proposed framework is described in Section II, and additional notations are expanded in Appendix A. The intermediary results on prior mean learning and the probabilistic control charts are presented and discussed in Section III and expanded in Appendix B and Appendix C. Comparison with other probabilistic MSPC methods is done in Appendix D.

II. MATERIALS AND METHODS

This paper extends the deterministic kernel-calibration framework introduced in Duma et al. [14] to a probabilistic setting. In the earlier work, kernel parameters were learned procedurally for K-MSPC using a supervised K-PCR-based intermediary objective, and the TEP benchmark was used for evaluation. In the present study, the focus shifts from deterministic point estimation to posterior uncertainty quantification and propagation. We therefore adopt the notation and deterministic K-PCA/K-MSPC framework [14] where possible, and introduce here only the additional elements required for Bayesian inference, probabilistic control charts, Gaussian-process-based calibration, and the unsupervised calibration route.

A. OVERVIEW OF THE PROPOSED PROBABILISTIC FRAMEWORK

The proposed methodology consists of three connected stages. The workflow is illustrated in Figure 1. First, kernel parameters are calibrated deterministically in order to obtain an informative reference point for inference. In the supervised route, this calibration is performed using labelled healthy/faulty data through either GPC or K-PCR-based discrimination. In the unsupervised route, only healthy-operation data are used and kernel parameters are selected using dataset-level structural criteria.

Second, the deterministic estimate is treated as the prior mean in a Bayesian formulation for the kernel parameters.

A posterior distribution is then inferred with MCMC, thereby quantifying epistemic uncertainty in the kernel lengthscale and variance parameters.Third, posterior draws of the kernel parameters are propagated through the K-PCA monitoring model. Each draw induces a corresponding K-PCA representation and therefore a corresponding pair of monitoring statistics, T^2 and SPE, together with variable contribution diagnostics. Aggregating over posterior draws yields probabilistic control charts and uncertainty-aware contribution plots.

The resulting framework may be interpreted as a probabilistic extension of deterministic K-MSPC: instead of a single calibrated monitoring model, one obtains a posterior distribution over monitoring models induced by uncertainty in the kernel parameters.

It is important to distinguish the proposed posterior propagation framework from other uncertainty-aware monitoring strategies, which are summarized in Table 1. Bootstrap-based MSPC estimates variability by repeatedly resampling the calibration data and refitting the monitoring model, which is useful for assessing sampling variability but does not provide a posterior distribution for the kernel parameters. Probabilistic PCA introduces uncertainty into a linear latent-variable model, whereas the present framework addresses uncertainty in the nonlinear kernel-induced feature-space geometry. Ensemble KPCA methods combine multiple KPCA submodels, often constructed using different kernel settings or local modelling structures, to improve nonlinear process monitoring robustness [20], [21]. However, such ensembles represent variability across a finite collection of candidate models rather than posterior uncertainty around a calibrated kernel configuration. Approximate kernel marginalisation is conceptually close to the present objective, but the MCMC formulation used here provides explicit posterior samples that can be propagated through the complete K-PCA monitoring pipeline. Consequently, each posterior draw induces a different kernel matrix, eigenspace, T^2 statistic, SPE statistic, and contribution profile, allowing uncertainty in both fault detection and fault attribution to be quantified.

Parameter selection is performed at three levels in the proposed workflow. First, kernel parameters are obtained through deterministic calibration. In the supervised route, the deterministic estimates are obtained using GPC/GPR-based and K-PCR-based intermediary objectives, while in the unsupervised route, several healthy-data criteria are evaluated to provide alternative kernel-parameter estimates. These deterministic estimates are not treated as final fixed quantities; instead, they define the prior means for the Bayesian inference stage. Second, the MCMC sampler used for posterior propagation is selected based on visual stationarity, posterior agreement, and effective sample size across the sampled kernel parameters. Third, the K-PCA monitoring settings, such as the number of retained components, are kept consistent across deterministic and probabilistic comparisons in order to isolate the effect of kernel-parameter uncertainty. In this way, the deterministic calibration stage provides optimized reference points, while the probabilistic stage quantifies how uncertainty around these selected parameters affects the resulting T^2, SPE, and contribution diagnostics.

B. BAYESIAN FORMULATION FOR KERNEL PARAMETER UNCERTAINTY

Let the calibration dataset be denoted by $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$, where $\mathbf{x}_i \in \mathbb{R}^p$ represents the i-th observation of an assumed gaussian multivariate distribution $\mathbf{X}$ and $y_i \in \{0, 1\}$ denotes an associated label indicating healthy or faulty operation when such labels are available.

We assume that the mapping between $\mathbf{x}_i$ and y_i depends on a parametric model, with unknown parameters $\boldsymbol{\theta} \in \mathbb{R}^m$.

Then, we are interested in estimating the posterior distribution of $\boldsymbol{\theta}$ given the dataset $\mathcal{D}$.

We work in the Bayesian framework, and the solution is given as a posterior distribution

$$p(\boldsymbol{\theta} \mid \mathcal{D}) \propto p(\mathcal{D} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta}), \tag{1}$$

where $p(\mathcal{D} \mid \boldsymbol{\theta})$ is the likelihood associated with the chosen intermediary model, and $p(\boldsymbol{\theta})$ is the prior distribution. We denote $\boldsymbol{\theta}$ as the kernel vector.

We consider two kernel families in this study, with the central modelling assumption that the kernel parameters should not be treated as fixed after calibration.

Instead, they are regarded as uncertain quantities inferred from finite process data. First we consider the isotropic squared exponential (SE), which has the form

$$k_{\mathrm{SE}}(\mathbf{x}_i, \mathbf{x}_j) = s_f^2 \exp\left(-\frac{1}{2}\frac{\|\mathbf{x}_i - \mathbf{x}_j\|^2}{\ell^2}\right), \tag{2}$$

where $\mathbf{x}_i$ is the i-th sample and $\mathbf{x}_j$ is the j-th sample. For the SE kernel, the kernel vector is

$$\boldsymbol{\theta}_{\mathrm{SE}} = \left[\ell, s_f, s_n\right],$$

where $\ell > 0$ is the global lengthscale, s_f^2 is the signal variance, and s_n^2 is the noise variance used in the intermediary likelihood. For the Automatic Relevance Determination (ARD) formulation,

$$k_{\mathrm{ARD}}(\mathbf{x}_i, \mathbf{x}_j) = s_f^2 \exp\left(-\frac{1}{2}\sum_{d=1}^{p}\frac{(x_{i,d} - x_{j,d})^2}{\ell_d^2}\right), \tag{3}$$

where $x_{i,d}$ is the i-th sample value in the d-th variable, and $x_{j,d}$ is the j-th sample value in the d-th variable. The kernel vector takes the form

$$\boldsymbol{\theta}_{\mathrm{ARD}} = \left[\ell_1, \ldots, \ell_p, s_f, s_n\right],$$

where each variable has its own characteristic lengthscale.

Choosing the prior distribution is crucial to the problem, as it is the only tuneable object after the kernel has been fixed. We choose weakly informative log-Gaussian models for the parameters, and for the prior mean, we use a deterministic estimate $\widehat{\boldsymbol{\theta}}$, obtained from either supervised or unsupervised calibration.

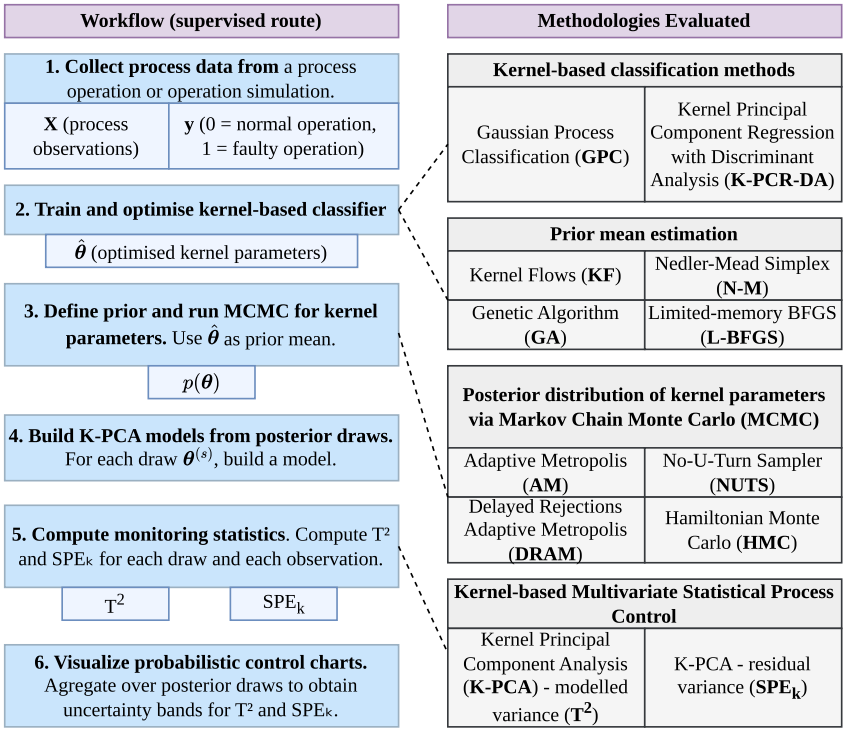


FIGURE 1. **Workflow for the generation of probabilistic control charts. Deterministic kernel calibration provides prior means, Bayesian sampling yields posterior draws of kernel parameters, and posterior propagation through K-PCA produces probabilistic monitoring statistics and contribution diagnostics. Here, is X a matrix of process observations (timeline) as rows, and process variables as columns, and y is the process state (either '0' if the process is in normal operation or '1' if the process is faulty).**

TABLE 1. **Positioning of the proposed probabilistic K-MSPC framework relative to alternative uncertainty-aware monitoring approaches.**

Approach	Main uncertainty source	Monitoring model affected	Relation to the proposed work
Deterministic K-MSPC	No explicit uncertainty; fixed calibrated kernel	Single K-PCA monitoring model	Standard baseline; kernel parameters are treated as fixed after calibration.
Bootstrap K-MSPC [51]	Sampling variability from resampled calibration data	Multiple refitted monitoring models	Captures data-resampling variability, but does not infer a posterior distribution of kernel parameters.
Probabilistic PCA monitoring [16]	Parameter and latent-variable uncertainty in a probabilistic linear model	Linear latent-variable monitoring model	Addresses uncertainty in linear PCA-type monitoring, not uncertainty in nonlinear kernel-induced geometry.
Probabilistic contribution analysis [17]	Diagnostic uncertainty under a probabilistic monitoring model	Contribution-based fault diagnosis	Quantifies diagnostic uncertainty, but not posterior uncertainty in K-MSPC kernel calibration.
Ensemble KPCA monitoring [20], [21]	Variability across multiple KPCA submodels or kernel settings	Finite collection of K-PCA monitoring models	Represents model-set variability, but not posterior uncertainty around calibrated kernel parameters.
Proposed probabilistic K-MSPC	Posterior uncertainty in calibrated kernel parameters	Distribution over K-PCA eigenspaces, T^2, SPE, and contribution plots	Directly propagates kernel-parameter uncertainty through nonlinear monitoring and diagnostics.

The deterministic kernel estimation is further presented in Section II-D. The priors are then

$$\log \ell \sim \mathcal{N}(\log \widehat{\ell}, \sigma^2_{\log \ell}),$$
$$\log s_f \sim \mathcal{N}(\log \widehat{s_f}, \sigma^2_{\log s_f}),$$
$$\log s_n \sim \mathcal{N}(\log \widehat{s_n}, \sigma^2_{\log s_n}).$$

We use analogous priors for the ARD lengthscales $\log(\ell_d)$. This parameterisation ensures positivity of the kernel quantities while allowing moderate posterior dispersion around the deterministic solution.

Two supervised and one unsupervised intermediary likelihood constructions are considered:

- **In the first supervised route**, kernel parameters are inferred through Gaussian process classification. A Gaussian process prior is placed on the latent function,

$$f(\mathbf{x}) \sim \mathcal{GP}\big(0, k_{\boldsymbol{\theta}}(\mathbf{x}, \mathbf{x}')\big),$$

where $k_{\boldsymbol{\theta}}$ is either k_{SE} or k_{ARD}. For computational convenience, the likelihood entering the posterior is written through the corresponding Gaussian-process

marginal form,

$$p(\mathcal{D} \mid \boldsymbol{\theta}) = \frac{1}{\sqrt{(2\pi)^n |\mathbf{C}_{\boldsymbol{\theta}}|}} \exp\left(-\frac{1}{2}\mathbf{y}^\top \mathbf{C}_{\boldsymbol{\theta}}^{-1}\mathbf{y}\right), \quad (4)$$

$$\mathbf{C}_{\boldsymbol{\theta}} = \mathbf{K}_{\boldsymbol{\theta}} + s_n^2 \mathbf{I}. \quad (5)$$

where $\mathbf{K}_{\boldsymbol{\theta}}$ is the kernel matrix induced by the current parameters. The associated log-likelihood is

$$\log p(\mathcal{D} \mid \boldsymbol{\theta}) = -\tfrac{1}{2}\mathbf{y}^\top \mathbf{C}_{\boldsymbol{\theta}}^{-1}\mathbf{y} - \tfrac{1}{2}\log|\mathbf{C}_{\boldsymbol{\theta}}| - \tfrac{n}{2}\log(2\pi) \quad (6)$$

- **In the second supervised route**, kernel parameters are inferred through K-PCR-based discrimination, extending the deterministic K-PCR calibration framework of Duma et al. [14] into a Bayesian setting. Let $\mathbf{T}_{\boldsymbol{\theta}} \in \mathbb{R}^{n\times r}$ denote the K-PCA score matrix induced by kernel parameters $\boldsymbol{\theta}$, and let $\hat{\mathbf{y}} = \mathbf{T}_{\boldsymbol{\theta}}\boldsymbol{\beta}$ denote a linear regression model in score space. Assuming Gaussian residuals,

$$\boldsymbol{\varepsilon} = \mathbf{y} - \mathbf{T}_{\boldsymbol{\theta}}\boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I}), \quad (7)$$

the likelihood becomes

$$p(\mathcal{D} \mid \boldsymbol{\theta}) = \mathcal{N}\left(\mathbf{y} \mid \mathbf{T}_{\boldsymbol{\theta}}\boldsymbol{\beta},\ \sigma^2\mathbf{I}\right) \propto \exp\left(-\tfrac{1}{2\sigma^2}\|\mathbf{y} - \mathbf{T}_{\boldsymbol{\theta}}\boldsymbol{\beta}\|^2\right). \quad (8)$$

- **In the unsupervised route**, no observed fault labels are initially available. Let $\mathcal{D} = \{(\mathbf{x}_i, \tilde{y}_i)\}_{i=1}^n$, where $\tilde{y}_i \in \{0, 1\}$ denotes the chart-induced pseudo-label assigned after deterministic calibration on healthy-operation data. Kernel parameters are first calibrated deterministically from healthy-operation data only, after which a provisional K-PCA monitoring model is constructed and observations are assigned induced labels $\tilde{y}_i \in \{0, 1\}$ according to whether their monitoring statistics remain within or exceed the chart limits. These induced labels are then used in the Bayesian stage through the same Gaussian-process marginal surrogate likelihood as in the first supervised route, namely

$$p(\mathcal{D} \mid \boldsymbol{\theta}) = \frac{1}{\sqrt{(2\pi)^n |\mathbf{C}_{\boldsymbol{\theta}}|}} \exp\left(-\frac{1}{2}\tilde{\mathbf{y}}^\top \mathbf{C}_{\boldsymbol{\theta}}^{-1}\tilde{\mathbf{y}}\right),$$
$$\log p(\mathcal{D} \mid \boldsymbol{\theta}) = -\tfrac{1}{2}\tilde{\mathbf{y}}^\top \mathbf{C}_{\boldsymbol{\theta}}^{-1}\tilde{\mathbf{y}} - \tfrac{1}{2}\log|\mathbf{C}_{\boldsymbol{\theta}}| - \tfrac{n}{2}\log(2\pi). \quad (9)$$

  The resulting posterior should therefore be interpreted as conditional on chart-induced pseudo-labels rather than externally observed fault classes.

In all three intermediary constructions, the posterior captures uncertainty in the kernel parameters induced by the calibration data and the chosen surrogate model. Posterior sampling is performed with four Markov chain Monte Carlo strategies: Adaptive Metropolis (AM) [35], Delayed Rejection Adaptive Metropolis (DRAM) [34], Hamiltonian Monte Carlo (HMC) [26], and the No-U-Turn Sampler (NUTS) [37]. Sampling is carried out in log-parameter scale, with chains initialised at the deterministic estimate $\widehat{\boldsymbol{\theta}}$. After burn-in removal, posterior summaries are computed from the retained draws.

Convergence and sampling efficiency are assessed using trace plots, autocorrelation functions, effective sample size (ESS) [38], and stability of posterior summaries across sampling windows. Agreement between different samplers is used as an additional qualitative robustness check.

C. PROBABILISTIC MONITORING STATISTICS AND POSTERIOR PROPAGATION

The deterministic K-PCA/K-MSPC machinery follows Duma et al. [14]; only its probabilistic extension is introduced here. Let $\boldsymbol{\theta}^{(m)}$, $m = 1, \ldots, M$, denote posterior draws from $p(\boldsymbol{\theta} \mid \mathcal{D})$. For each draw, a kernel matrix is constructed using either k_{SE} or k_{ARD}, a K-PCA model is obtained, and the corresponding monitoring statistics are evaluated for each sample.

Thus, each posterior draw induces a draw-specific pair of monitoring statistics,

$$T^{2\,(m)}(\mathbf{x}), \quad \mathrm{SPE}^{(m)}(\mathbf{x}),$$

as well as draw-specific contribution diagnostics,

$$C_d^{T^2,(m)}(\mathbf{x}), \quad C_d^{\mathrm{SPE},(m)}(\mathbf{x}).$$

The posterior sample $\{T^{2\,(m)}(\mathbf{x})\}_{m=1}^M$ defines an empirical posterior distribution for the T^2 statistic at sample $\mathbf{x}$, and similarly for SPE and the contribution terms. Point summaries such as posterior means or medians can therefore be plotted as representative control charts, while credible intervals quantify the uncertainty induced by kernel parameter variability.

In this work, probabilistic control charts are constructed by summarising the posterior distribution of the monitoring statistics at each time point. In the same manner, uncertainty-aware contribution plots are obtained by summarising the posterior distribution of the contribution coefficients for each variable. This allows not only fault detection under uncertainty, but also uncertainty quantification of variable attribution.

An important conceptual point is that the uncertainty considered here is structural rather than measurement-level noise around a fixed model. Each posterior draw modifies the kernel geometry, which in turn changes the K-PCA eigenspace, the monitored subspace/residual decomposition, and consequently the values of T^2, SPE, and their contributions. The probabilistic control charts therefore reflect uncertainty in the monitoring model itself.

D. DETERMINISTIC KERNEL CALIBRATION ROUTES

The role of deterministic calibration in the present study is to provide informative prior means for the Bayesian stage. Three calibration routes are considered: supervised calibration through GPC, supervised calibration through K-PCR, and unsupervised calibration from healthy-operation data only.

1) SUPERVISED CALIBRATION FOR PROCESSES WITH KNOWN FAULT INFORMATION

The first supervised route uses Gaussian process classification to distinguish healthy and faulty observations. Kernel parameters are estimated by deterministic optimisation of the intermediary objective with respect to $\boldsymbol{\theta}$. The optimisers considered are L-BFGS [40], Nelder–Mead [32], Genetic Algorithm [31], and Kernel Flows [43]. Optimisation is performed in log-parameter scale to preserve positivity of the kernel quantities. This route is new relative to Duma et al. [14] and is included here because it provides an alternative supervised surrogate for kernel calibration before posterior inference.

The second supervised route follows the deterministic K-PCR-based calibration philosophy introduced in Duma et al. [14]. In that framework, kernel parameters are selected through discrimination performance in the same truncated K-PCA representation used later for monitoring, making the calibration structurally aligned with K-MSPC. In the present paper, those deterministic K-PCR estimates are not an end point; instead, they are used as prior means for subsequent Bayesian sampling. Because this deterministic K-PCR framework has already been presented in detail, it is not re-derived here.

2) UNSUPERVISED CALIBRATION FROM HEALTHY-OPERATION DATA

In many process-monitoring applications, labelled fault data are unavailable. To address this setting, an unsupervised calibration route is included in the present study. Here, kernel parameters are inferred exclusively from healthy-operation data before Bayesian posterior inference.

For the unsupervised route, calibration is restricted to the isotropic SE kernel with a single global lengthscale ℓ. This choice favours parsimony and robustness, and avoids the severe ill-posedness that would arise when estimating many variable-specific lengthscales without supervision. Ten unsupervised selection rules are considered, each targeting a different structural property of the healthy-data kernel representation. These are summarised in Table 2.

For this unsupervised route, the workflow differs from the supervised case only in the source of the prior mean. Normal-operation data are first used to select the deterministic kernel parameter. A K-PCA monitoring model is then fitted, control limits are computed, and new observations are provisionally mapped into in-control/out-of-control classes through their position relative to the chart limits. These induced labels are then used in the subsequent Bayesian sampling stage. The unsupervised workflow is illustrated in Figure 2.

The notation and deterministic K-PCA/K-MSPC framework follow Duma et al. [14], including kernel centering, score computation, construction of T^2 and SPE control charts, and contribution diagnostics. To avoid repeating material from the earlier paper, these definitions are summarised in Appendix A. The present paper introduces only the additional notation associated with kernel parameter posteriors, MCMC sampling, and posterior propagation.

E. COMPUTATIONAL COMPLEXITY AND PRACTICAL IMPLEMENTATION

The proposed probabilistic K-MSPC framework introduces additional computational cost compared with deterministic K-MSPC because the K-PCA monitoring model is reconstructed for multiple posterior draws of the kernel parameters. Let n denote the number of healthy calibration observations, p the number of process variables, r the number of retained K-PCA components, M the number of posterior samples propagated through the monitoring model, and n_{te} the number of test observations.

For a single kernel-parameter value, constructing the full kernel matrix requires approximately $\mathcal{O}(n^2p)$ operations for distance-based kernels such as the squared exponential kernel. The subsequent K-PCA decomposition has cost $\mathcal{O}(n^3)$ if a full eigendecomposition is performed. In practice, only the leading r components are required for monitoring, and iterative eigensolvers can reduce the eigendecomposition cost to approximately $\mathcal{O}(n^2r)$ after the kernel matrix has been constructed. Projection of n_{te} new observations requires construction of the test–training kernel matrix, with cost approximately $\mathcal{O}(n_{\text{te}}np)$, followed by score and monitoring-statistic computation.

In the posterior propagation stage, this reconstruction is repeated for M posterior draws. Therefore, when truncated eigendecomposition is used, the dominant cost of posterior propagation scales approximately as

$$\mathcal{O}\left(M(n^2p + n^2r + n_{\text{te}}np)\right). \tag{10}$$

If full eigendecomposition is used, the corresponding cost becomes approximately

$$\mathcal{O}\left(M(n^2p + n^3 + n_{\text{te}}np)\right). \tag{11}$$

This shows that the number of calibration observations and the number of propagated posterior draws are the main drivers of computational cost.

The computational burden can be reduced in several ways. First, posterior propagation is embarrassingly parallel because each posterior draw can be processed independently. Second, posterior thinning or selecting a smaller representative subset of posterior samples can reduce M while preserving the main posterior variability. Third, for large-scale industrial datasets, low-rank kernel approximations such as Nyström approximations or random Fourier features can be used to avoid repeated construction and decomposition of full $n \times n$ kernel matrices. Finally, in online applications, the computationally expensive posterior sampling and K-PCA reconstruction can be performed offline, while real-time monitoring only requires projection of new observations through a selected set of posterior monitoring models.

TABLE 2. Unsupervised kernel-parameter tuning methods evaluated for deterministic K-MSPC.

Method	Description	Inspired by
M1 Median all-pairs	RBF lengthscale set to the median of all pairwise distances computed from healthy operation data.	[33], [47]
M2 Median kNN	Lengthscale set to the median of k-nearest neighbour distances, improving robustness of local similarity measures to outliers and nonstationarity.	[36], [53]
M3 Target mean similarity	Kernel lengthscale chosen such that the average off-diagonal kernel similarity matches a predefined target value.	[48]
M4 Effective rank target	Kernel tuned to achieve a desired effective rank (spectral entropy) of the centered Gram matrix, controlling model complexity.	[24], [45]
M5 Split-half subspace	Lengthscale selected to maximise stability of K-PCA subspaces across random splits of healthy data.	[39]
M6 Quantile stability	Kernel selected to minimise variability of healthy T^2 and SPE control limits across resampled datasets.	[42]
M7 One-class SVM proxy	Kernel tuned using one-class SVM criteria, such as support fraction and rejection stability, as a proxy objective.	[46], [52]
M8 MMD halves	Lengthscale chosen to minimise the maximum mean discrepancy between temporal halves of healthy operation data.	[27], [33]
M9 KRR self-predict	Kernel ridge regression trained to reconstruct healthy data; lengthscale minimises cross-validated prediction error.	[44], [48]
M10 Alignment kNN graph	Kernel aligned with a k-nearest-neighbour graph to preserve local manifold structure of healthy data.	[25], [29]

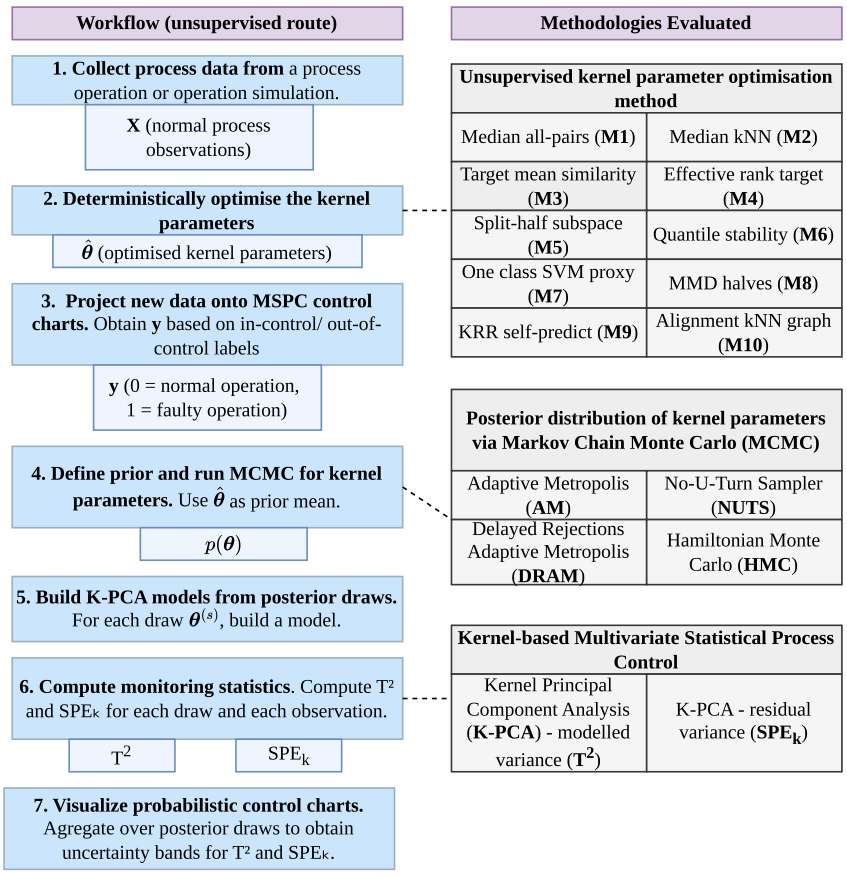


FIGURE 2. Workflow for generation of probabilistic control charts when no fault data are initially available. Deterministic unsupervised kernel calibration is followed by provisional chart-based label assignment and subsequent Bayesian posterior inference.

F. CASE STUDY: TENNESSEE EASTMAN PROCESS

The proposed framework is evaluated on the Tennessee Eastman Process benchmark, following the same general process setting as in Duma et al. [14]. The TEP is a widely used nonlinear industrial process simulation with interacting process and manipulated variables and 21 predefined fault

TABLE 3. Approximate computational complexity of the main stages of the proposed probabilistic K-MSPC framework. Here, n is the number of calibration observations, p is the number of variables, r is the number of retained K-PCA components, M is the number of propagated posterior draws, and n_{te} is the number of test observations.

Stage	Approximate cost	Main cost driver
Kernel matrix construction	$\mathcal{O}(n^2p)$	Pairwise distances between calibration observations.
Full K-PCA eigendecomposition	$\mathcal{O}(n^3)$	Full decomposition of the centered kernel matrix.
Truncated K-PCA eigendecomposition	$\mathcal{O}(n^2r)$	Leading r eigenvectors/eigenvalues needed for monitoring.
Projection of test observations	$\mathcal{O}(n_{te}np)$	Test–training kernel matrix construction.
Posterior propagation with truncated K-PCA	$\mathcal{O}\left(M(n^2p + n^2r + n_{te}np)\right)$	Repeating K-PCA reconstruction and projection for M posterior draws.
Posterior propagation with full K-PCA	$\mathcal{O}\left(M(n^2p + n^3 + n_{te}np)\right)$	Repeating full eigendecomposition for M posterior draws.

scenarios. Detailed process and variable descriptions are available in [14], [28], and [30].

In the present work, healthy-operation data are used to construct the monitoring model and to support unsupervised calibration, while healthy/faulty labelled data are additionally used in the supervised intermediary calibration routes. Independent test partitions with fault onset during the monitoring trajectory are used to evaluate deterministic and probabilistic chart performance. This setup preserves comparability with the earlier deterministic study while allowing the new probabilistic framework to be assessed under both supervised and unsupervised calibration conditions.

III. RESULTS AND DISCUSSION

A. SUPERVISED CALIBRATION

1) SUPERVISED INTERMEDIARY STEP: KERNEL-BASED CLASSIFICATION

Figure 3 summarizes deterministic parameter optimisation for the GPC model with an SE kernel (fault **F01**). All four optimisers converged to solutions yielding perfect fault detection in the subsequent control charts. Among the tested methods, L-BFGS was the most computationally efficient, requiring 2.775 s until automatic convergence, whereas the GA was substantially slower (99.330 s).

The convergence behaviour differs across optimisers (Fig. 3a–3b). GA and Nelder–Mead converged to similar kernel lengthscale values. Kernel Flows exhibits a smooth initial trajectory, followed by noticeable oscillations near convergence. Nelder–Mead shows a characteristic stepwise evolution, with the lengthscale progressing through several plateaus before settling to the final value.

Although all optimisation methods achieved strong convergence in the classification setting, as seen in Table 4, good classification performance does not necessarily translate into optimal behaviour in the subsequent control charts. This highlights the distinction between kernel calibration for supervised discrimination and calibration for statistical process monitoring.

In previous work [14], Kernel Flows was shown to be an effective strategy for learning kernel parameters in the context of Kernel MSPC via K-PCR. However, for GPC, the results in Table 4 indicate a slightly different ranking. Based on the mean performance across faults, the overall ordering is GA > N-M > L-BFGS > KF. Nevertheless, the performance differences between GA, N–M, and L-BFGS are marginal.

Given the substantial computational cost of GA (approximately 60 seconds) compared to L-BFGS (approximately 0.6 seconds), the small improvement in average performance does not justify the additional computational burden in the GPC setting. Consequently, L-BFGS was selected as the preferred optimiser for the supervised calibration stage using GPC as a classifier. It is also worth noting that many fault results are identical across optimisation methods. Only a few cases (notably **F13**, **F15**, **F16**, and **F18**) show slight improvements for Nelder–Mead and GA compared to L-BFGS. Overall, the results suggest that first-order gradient-based optimisation provides an efficient and sufficiently accurate solution for kernel parameter estimation in the GPC-based intermediary step.

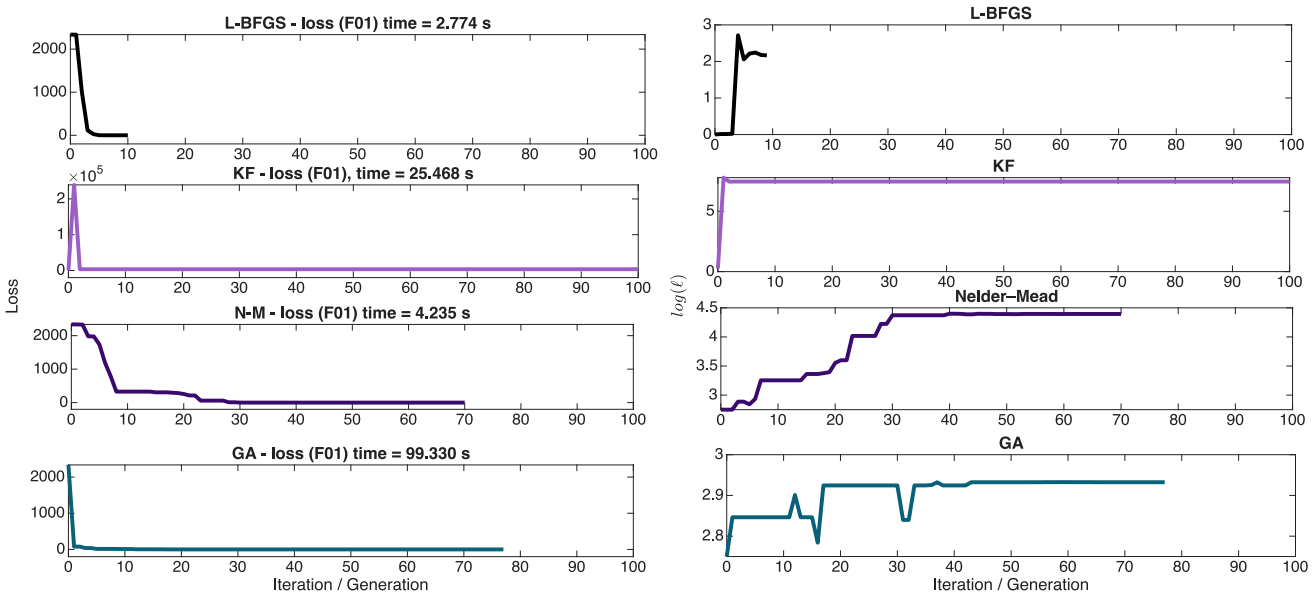
2) DETERMINISTIC CONTROL CHARTS: SUPERVISED CALIBRATION

For Fault F01, although perfect classification performance was achieved in the supervised GPC stage, the corresponding control-chart behaviour reveals important differences. In particular, convergence of the kernel parameters for the classifier does not imply equivalent performance when the same parameters are used in a K-PCA–based MSPC framework.

When applying K-PCA MSPC with kernel parameters optimised via the GPC–L-BFGS route, the resulting deterministic monitoring statistics (Figure 4a-b) detect the onset of the fault correctly. The contribution analysis for a sample taken 80 minutes after the fault start (Figure 4c-d) provides further insight. Variables $\mathbf{x}_{16}$, $\mathbf{x}_1$ and $\mathbf{x}_{44}$ are the primary drivers of the out-of-control behaviour in the T^2 statistic, indicating that the dominant deviations lie within the principal subspace captured by the K-PCA model. Variable $\mathbf{x}_{16}$ presents also new variation, as seen in the contribution chart for explaining the SPE_k value at timestamp 180.

3) MCMC RESULTS

Figure 5 presents the MCMC chains obtained for the SE kernel using four sampling strategies: AM, DRAM, HMC, and the NUTS. The prior means were set to the deterministic L-BFGS estimates from the GPC optimisation stage.



(a) (b)

FIGURE 3. **Deterministic kernel parameter optimization (a) loss trace and (b) kernel parameter convergence for fault F01. All four converged in a perfect fault detection in the control charts.**

TABLE 4. **GPC optimization with an SE Kernel. Performance comparison of four parameter optimisation methods (LBFGS, Kernel Flows, Nelder–Mead, Genetic Algorithm) across 21 TEP faults. Metrics include AUC, F1-score, False Alarm Rate (FAR), and Fault Detection Rate (FDR).**

Fault	L-BFGS				Kernel Flows				Nelder-Mead				Genetic Algorithm			
	AUC	F1	FAR	FDR	AUC	F1	FAR	FDR	AUC	F1	FAR	FDR	AUC	F1	FAR	FDR
F01	1.00	1.00	0.00	1.00	1.00	1.00	0.00	1.00	1.00	1.00	0.00	1.00	1.00	1.00	0.00	1.00
F02	0.99	0.98	0.00	0.96	0.99	0.98	0.00	0.95	0.99	0.98	0.00	0.97	0.99	0.98	0.00	0.97
F03	0.49	0.58	0.46	0.45	0.44	0.00	0.00	0.00	0.50	0.63	0.54	0.51	0.50	0.64	0.55	0.53
F04	1.00	1.00	0.01	1.00	0.50	0.00	0.00	0.00	1.00	1.00	0.00	1.00	1.00	1.00	0.00	1.00
F05	1.00	1.00	0.00	1.00	0.39	0.00	0.00	0.00	1.00	1.00	0.00	1.00	1.00	1.00	0.00	1.00
F06	1.00	1.00	0.00	1.00	1.00	0.91	0.00	0.83	1.00	1.00	0.00	1.00	1.00	1.00	0.00	1.00
F07	1.00	1.00	0.00	1.00	1.00	1.00	0.00	1.00	1.00	1.00	0.00	1.00	1.00	1.00	0.00	1.00
F08	0.99	0.86	0.00	0.75	0.91	0.26	0.00	0.15	0.99	0.88	0.00	0.78	0.99	0.88	0.00	0.78
F09	0.68	0.54	0.15	0.38	0.60	0.00	0.00	0.00	0.66	0.57	0.18	0.41	0.66	0.57	0.18	0.41
F10	0.81	0.73	0.18	0.60	0.24	0.00	0.00	0.00	0.84	0.80	0.19	0.70	0.84	0.80	0.19	0.70
F11	0.72	0.62	0.18	0.46	0.37	0.00	0.00	0.00	0.74	0.69	0.19	0.54	0.74	0.69	0.19	0.55
F12	0.98	0.88	0.00	0.79	0.98	0.88	0.00	0.78	0.99	0.88	0.00	0.78	0.99	0.87	0.00	0.78
F13	0.81	0.28	0.00	0.16	0.78	0.17	0.01	0.09	0.79	0.63	0.00	0.46	0.79	0.63	0.00	0.46
F14	0.99	0.97	0.03	0.94	0.48	0.00	0.00	0.00	1.00	0.97	0.03	0.96	1.00	0.97	0.03	0.96
F15	0.68	0.53	0.12	0.37	0.39	0.00	0.00	0.00	0.70	0.63	0.14	0.48	0.70	0.64	0.14	0.49
F16	0.78	0.77	0.22	0.65	0.40	0.00	0.00	0.00	0.82	0.83	0.28	0.75	0.82	0.83	0.28	0.75
F17	0.95	0.86	0.03	0.76	0.35	0.00	0.00	0.00	0.95	0.88	0.03	0.78	0.95	0.88	0.03	0.79
F18	0.38	0.39	0.00	0.24	0.24	0.03	0.03	0.01	0.51	0.52	0.00	0.35	0.52	0.52	0.00	0.35
F19	0.55	0.78	0.84	0.75	0.27	0.17	0.01	0.09	0.63	0.80	0.76	0.78	0.70	0.81	0.63	0.77
F20	0.88	0.75	0.02	0.60	0.24	0.00	0.00	0.00	0.88	0.76	0.03	0.61	0.88	0.76	0.03	0.62
F21	0.48	0.22	0.06	0.13	0.32	0.00	0.00	0.00	0.49	0.23	0.08	0.13	0.49	0.23	0.08	0.13
Mean	0.82	0.75	0.11	0.67	0.57	0.26	0.00	0.23	0.83	0.79	0.12	0.71	0.84	0.80	0.11	0.72

In terms of convergence behaviour, AM and DRAM exhibit similar posterior trajectories and stabilise around comparable parameter regions. Likewise, HMC and NUTS converge to closely aligned posterior distributions, reflecting their shared gradient-based sampling structure. While all samplers reach stationarity, the gradient-based methods demonstrate smoother exploration of the parameter space. For the kernel lengthscale parameter $\log(\ell)$, all four samplers converge to posterior modes that are larger than the prior mean, and in a similar MAP value. This indicates that the probabilistic calibration favours a smoother kernel (i.e., larger lengthscale) than suggested by the deterministic optimisation alone. For the signal variance parameter $\log(s_f)$, a clear distinction is observed between sampling families. NUTS and

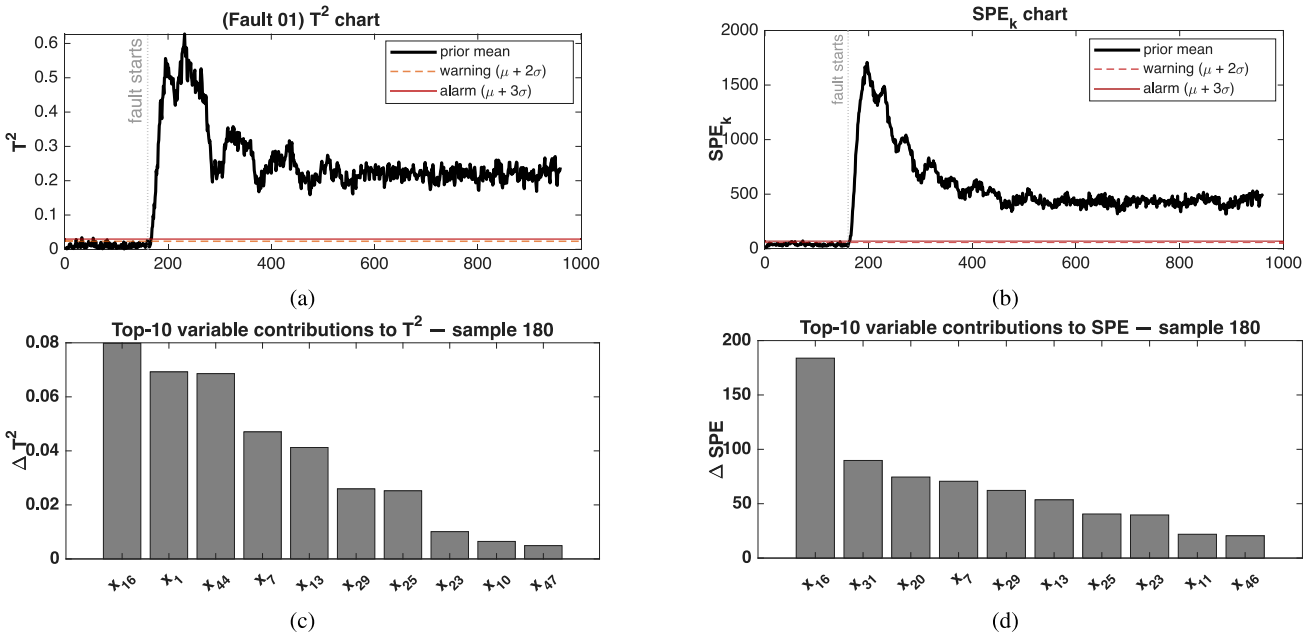


FIGURE 4. K-PCA MSPC control charts calibrated with the GPC-L-BFGS convergence kernel parameters, for a SE kernel.

DRAM converge to posterior values that are slightly smaller than the prior mean, suggesting a reduction in effective signal amplitude when uncertainty is fully accounted for. In contrast, AM yields posterior estimates that remain close to the prior mean, indicating weaker adjustment of this parameter. Finally, for the noise variance parameter $\log(s_n)$, all four sampling approaches converge near the prior mean. This suggests that the deterministic estimate already provides a stable and well-identified value for the noise level, with limited posterior correction required.

To make the sampler comparison systematic, the four MCMC strategies were evaluated using three criteria: visual stationarity of the post-burn-in chains, agreement of the posterior distributions, and effective sample size (ESS) for each sampled parameter. Since posterior propagation through K-PCA depends on the full kernel-parameter vector, the minimum ESS across parameters was used as the main practical criterion for selecting the sampler used in the subsequent probabilistic control charts. This criterion is conservative, because poor mixing in any single kernel parameter can affect the reconstructed kernel matrix, eigenspace, and monitoring statistics.

ESS was computed for each parameter using the post-burn-in chains. The results indicate that DRAM achieves the highest and most balanced sampling efficiency, with ESS values between approximately 305 and 486 across parameters. The AM sampler produces moderately efficient chains, with ESS values around 188–223. In contrast, the gradient-based samplers show more uneven efficiency under the current tuning settings. For HMC, the ESS values for the kernel parameters $\log(\ell)$ and $\log(s_f)$ are relatively low, approximately 18, indicating strong autocorrelation in these chains, although the noise variance parameter $\log(s_n)$ mixes efficiently with ESS = 833. NUTS improves upon HMC but still yields lower ESS values for $\log(\ell)$ and $\log(s_f)$, approximately 41, compared with the adaptive random-walk samplers.

Based on these diagnostics, DRAM was selected for the subsequent posterior propagation experiments. This choice was motivated by its stable posterior trajectory, agreement with the other converged samplers, and the largest minimum ESS across the kernel parameters. Although HMC and NUTS reached stable posterior regions rapidly, their lower ESS for the kernel parameters indicates stronger autocorrelation under the present tuning configuration. Therefore, DRAM was considered the most reliable sampler for representing the posterior uncertainty propagated to the probabilistic T^2, SPE, and contribution plots.

Considering the minimum ESS across parameters, DRAM provides the most efficient exploration of the posterior distribution, followed by AM, while HMC and NUTS exhibit slower mixing for the kernel parameters. These results suggest that, for this posterior geometry, adaptive Metropolis-type samplers provide more robust sampling performance than the gradient-based methods under the current tuning settings.

The MCMC results for Kernel PCR are presented in Appendix C. In contrast to the GPC-based optimisation, the AM and DRAM samplers no longer converge to the same posterior distribution of the kernel parameters. Instead, DRAM, HMC, and NUTS converge to posterior distributions centred around a similar value of $\log(\ell)$, whereas AM explores a slightly different region of the parameter space.

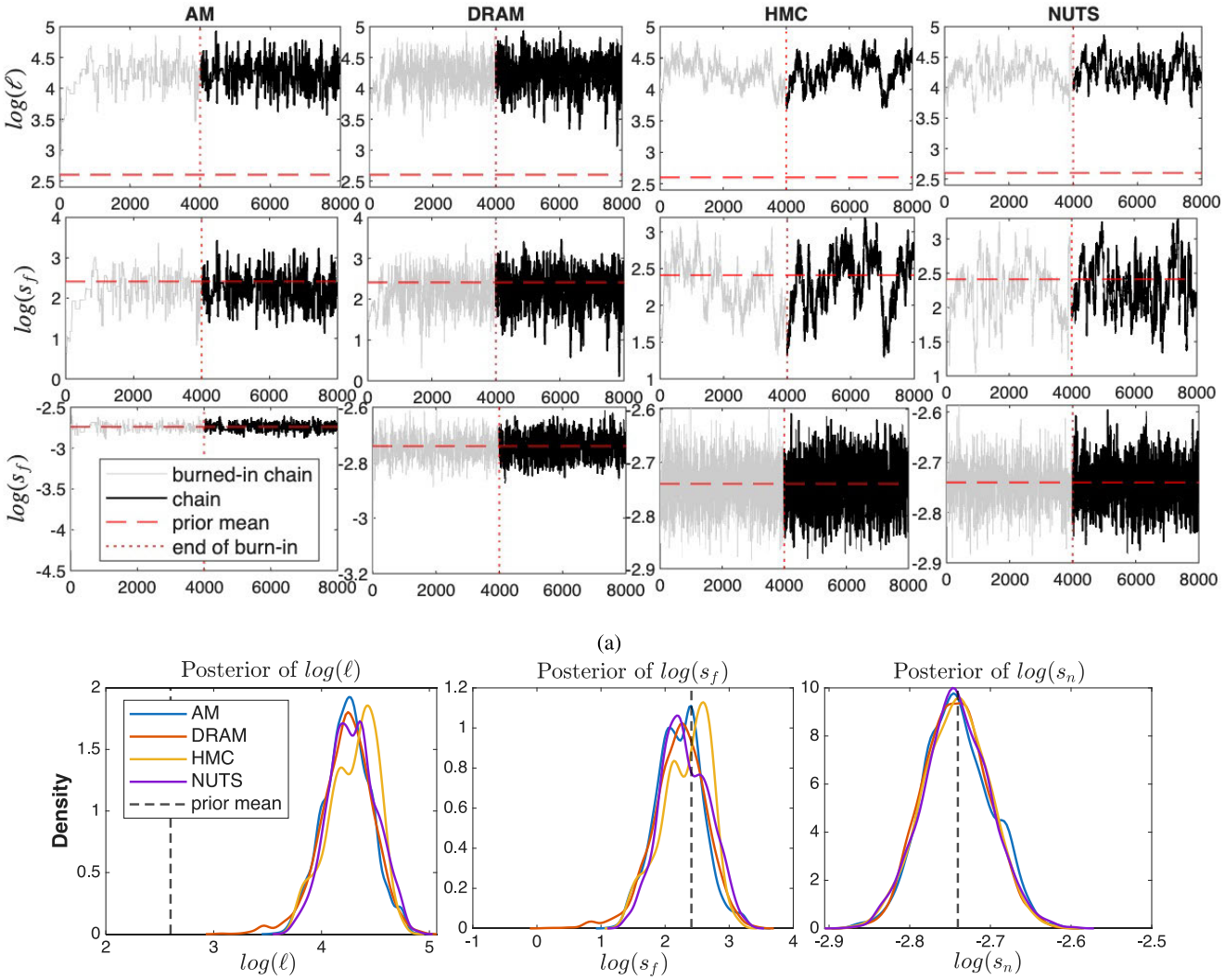


FIGURE 5. MCMC comparison for the GPC-based log-likelihood using the L-BFGS-GPC kernel estimates as prior means: (a) post-burn-in chain behaviour and (b) posterior distributions for the sampled log-kernel parameters. The samplers were compared using visual stationarity, agreement of posterior distributions, and effective sample size (ESS). DRAM was selected for posterior propagation because it provided the most balanced sampling efficiency, with the largest minimum ESS across parameters, while remaining consistent with the posterior regions identified by the other converged samplers.

In terms of convergence speed, HMC exhibits the fastest stabilisation, reaching a stationary regime after approximately 100 iterations. NUTS converges shortly thereafter, around 250 iterations, while DRAM requires a longer burn-in period of roughly 1000 iterations before stabilising. This highlights the improved efficiency of gradient-based samplers for the K-PCR likelihood landscape.

4) PROBABILISTIC MSPC CONTROL CHARTS: SUPERVISED CALIBRATION

Figures 6 and 7 present the probabilistic T^2 and SPE control charts, together with their corresponding contribution plots, for the SE and ARD-SE kernels, respectively. A clear pattern is observed in the uncertainty bands. When the process operates within control limits (healthy region), the posterior uncertainty in both T^2 and SPE is very small, indicating high confidence in the in-control state. However, once the fault is initiated, the uncertainty increases substantially. This reflects the epistemic uncertainty in kernel parameters being amplified under abnormal process conditions.

Moreover, the uncertainty is consistently larger for the SPE control chart than for the T^2 chart. This behaviour is expected, as SPE captures residual variation outside the modeled K-PCA subspace and is therefore more sensitive to model misspecification and parameter variability. In contrast, T^2 represents variation within the retained principal component subspace, which is typically more stable. The same behaviour is observed in the contribution plots. Variable contributions to the SPE chart exhibit noticeably wider uncertainty bands than those associated with the T^2 chart. This indicates that uncertainty in kernel parameters

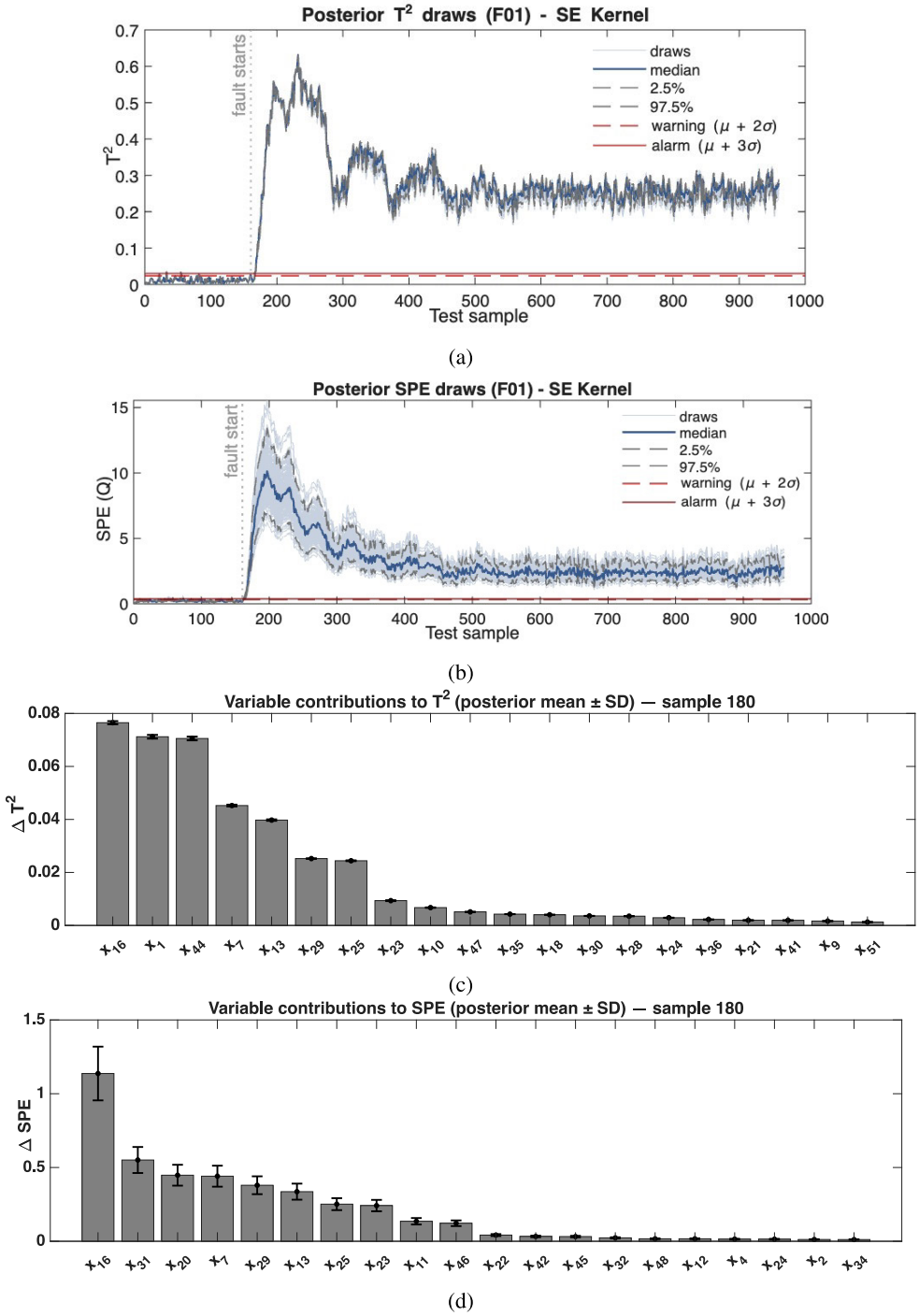


FIGURE 6. Probabilistic control charts (a,b) and variable contributions to the control charts (c, d) for the SE Kernel K-PCA, with DRAM MCMC sampling and GPC-based prior mean.

propagates more strongly to residual-space diagnostics than to subspace-based variation.

A relationship between contribution magnitude and uncertainty can be observed: variables with lower contribution values tend to exhibit smaller posterior uncertainty, whereas variables driving the out-of-control behaviour display both larger contributions and wider uncertainty intervals. This suggests that uncertainty propagation is proportional to the strength of a variable's influence on the monitoring statistic. Overall, the probabilistic formulation not only improves fault detection performance but also provides additional interpretability by quantifying confidence in both detection and variable attribution.

In Appendix B, Table 6 compares the fault detection performance of the deterministic control charts (prior mean of the kernel parameters) with the posterior-mean-based charts

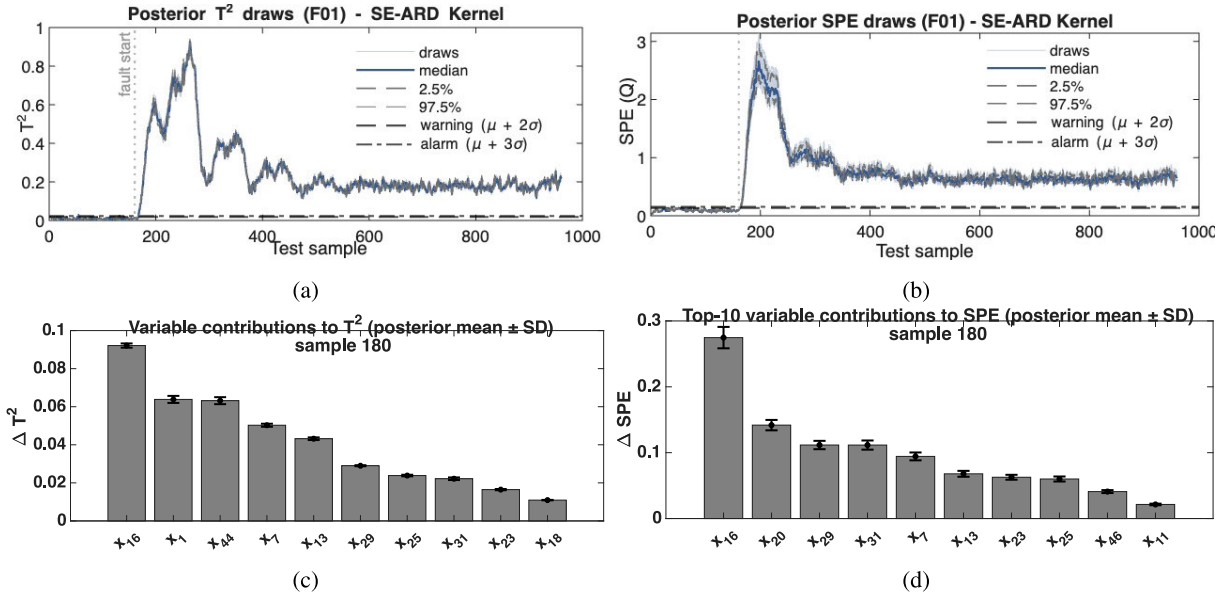


FIGURE 7. Probabilistic control charts (a,b) and variable contributions to the control charts (c, d) for the SE-ARD kernel, with DRAM sampling and GPC-based prior mean. Similar visualisations for K-PCR are found in Appendix C.

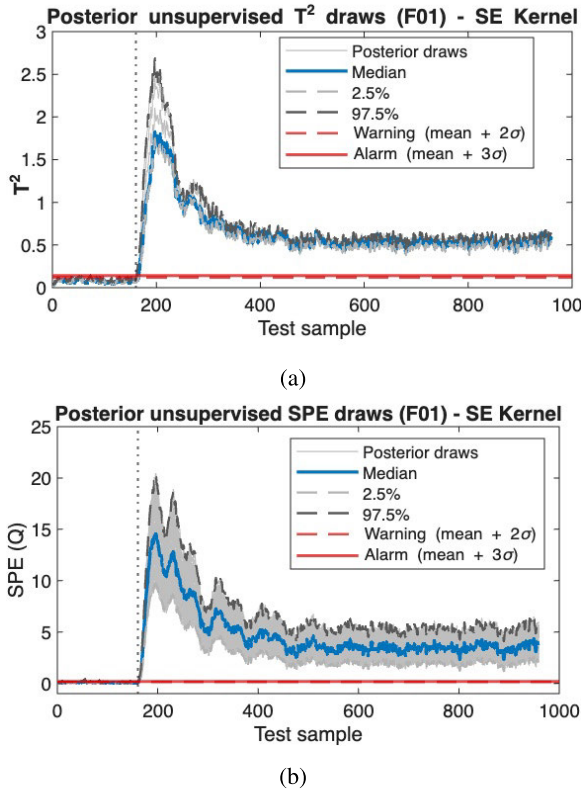


FIGURE 8. Probabilistic control charts for (a) T2 and (b) SPE metrics, using unsupervised kernel calibration (method M8).

for the SE kernel. The comparison is made in terms of the composite indicator (CI), which summarises the trade-off between false alarm rate (FAR) and fault detection rate (FDR).

The results show that nearly all faults benefit from the probabilistic treatment of the kernel parameters. In several cases, the improvement is substantial. For instance, Fault **F01** improves from CI = 0.82 to 0.99, **F04** from 0.57 to 0.75, **F14** from 0.53 to 0.92, **F17** from 0.60 to 0.89, **F20** from 0.61 to 0.71, and **F21** from 0.50 to 0.81. These results indicate that incorporating posterior uncertainty into the control chart construction can significantly enhance fault detectability, particularly for faults that are only partially captured under the deterministic (prior-mean) setting.

When employing the ARD-SE kernel (Figures 7), several noteworthy differences can be observed in comparison with the isotropic SE kernel. The overall uncertainty in both the T^2 and SPE control charts is reduced, while the qualitative uncertainty pattern remains unchanged: uncertainty is small during normal operation and increases after the onset of the fault. This suggests that the larger posterior variability is primarily associated with abnormal operating conditions rather than being an artifact of kernel choice.

For the T^2 contribution plots, the ranking of the most influential variables remains largely unchanged, although their relative magnitudes are modified. In particular, variable $\mathbf{x}_{16}$ becomes the dominant contributor to the out-of-control behaviour. In the SPE contribution charts, the posterior uncertainty is again reduced compared to the isotropic SE case, while minor changes in the top-10 ranking of contributing variables can be observed. This indicates that residual-space diagnostics are more sensitive to anisotropic scaling, as ARD modifies how unexplained variation is allocated across variables.

With respect to the ARD-SE kernel, the comparison between deterministic prior-mean and posterior-mean control charts is presented in Table 7 in Appendix B. In contrast to

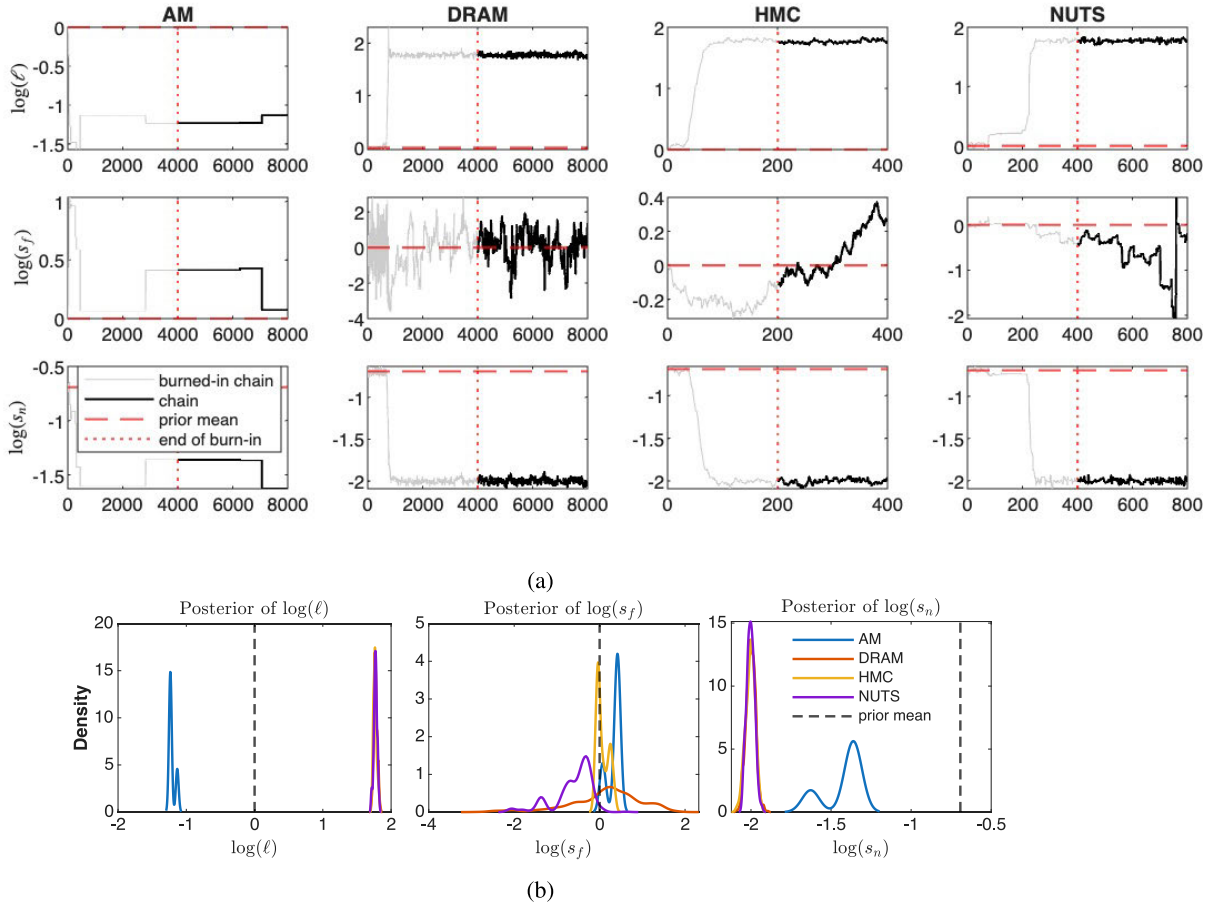


FIGURE 9. MCMC (a) chains for the K-PCR-based log-likelihood, with N-M convergence values as prior mean and (b) the posterior distribution of the SE kernel parameters.

the isotropic SE kernel case, only marginal differences are observed between the deterministic and probabilistic ARD-SE approaches. That is, incorporating posterior uncertainty in the ARD-SE setting does not lead to substantial additional improvements in the composite indicator.

The reduction in posterior uncertainty observed for the ARD-SE kernel can be interpreted through its variable-specific lengthscales. In the isotropic SE kernel, a single global lengthscale controls similarity across all process variables. This global parameter must therefore compromise between variables that may differ in scale, noise level, process relevance, and sensitivity to the fault. Small posterior changes in this shared lengthscale can consequently alter the full kernel matrix and the resulting K-PCA eigenspace, leading to wider uncertainty bands in the monitoring statistics.

In contrast, the ARD-SE kernel assigns an individual lengthscale to each process variable. Variables that are more relevant for representing the process state can be associated with shorter lengthscales and therefore stronger influence on the kernel similarity, whereas less informative or noisier variables can be associated with longer lengthscales and become effectively down-weighted. This variable-specific scaling reduces the need for a single global lengthscale to represent all directions of variation simultaneously. As a result, posterior perturbations in individual lengthscales have a more localized effect on the kernel geometry, which can stabilize the K-PCA eigenspace and reduce the propagated uncertainty in T^2, SPE, and contribution diagnostics.

This interpretation is consistent with the observation that the deterministic and probabilistic ARD-SE results are closer to each other than in the isotropic SE case. The additional flexibility of ARD appears to absorb part of the structural uncertainty at the deterministic calibration stage, leaving less posterior variability to be propagated through the monitoring model. However, this should not be interpreted as ARD-SE being uniformly preferable in all settings; rather, its benefit depends on whether the process variables have heterogeneous relevance and scaling with respect to the monitored fault behaviour. An exception in the present study is fault **F21**, for which the isotropic SE configuration yields better performance than the ARD-SE setup.

B. UNSUPERVISED CALIBRATION

Table 5 presents the composite indicator results for the unsupervised deterministic kernel calibration, where only the SE kernel parameters were optimised. With the exception

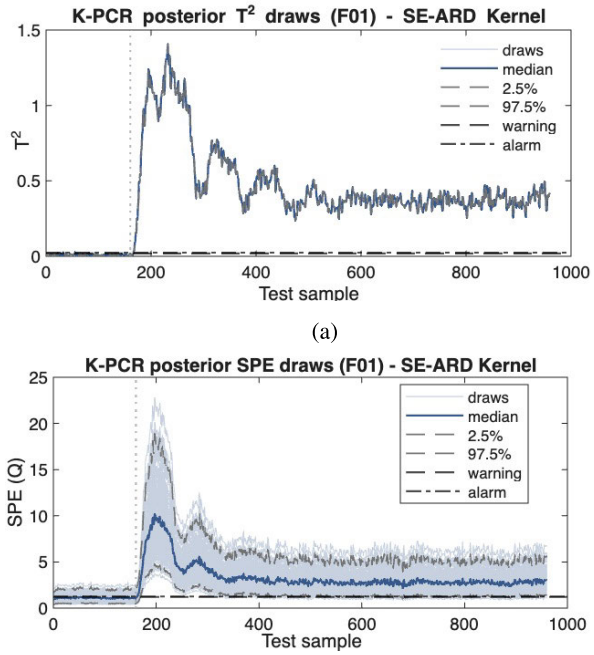
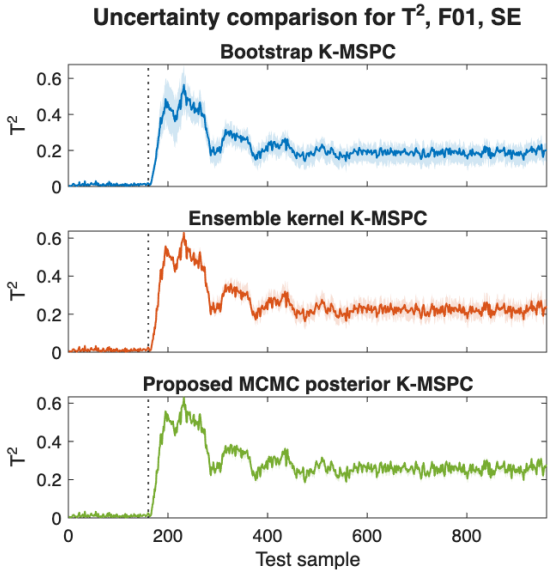
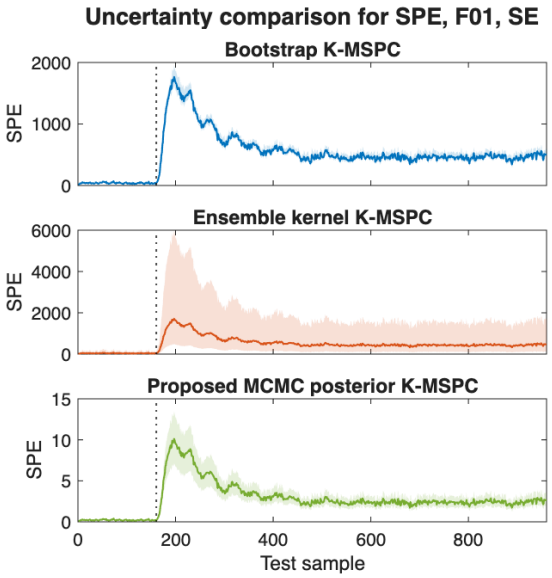


FIGURE 10. The probabilistic control charts for the F01 fault, using the SE-ARD kernel in a K-PCR prior mean estimation set-up.



(a) T^2 probabilistic control chart comparison.



(b) SPE probabilistic control chart comparison.

FIGURE 11. Comparison of uncertainty-aware control charts for a representative fault scenario. The proposed MCMC posterior K-MSPC exhibits narrower uncertainty bands while preserving fault detectability relative to bootstrap K-MSPC and ensemble kernel K-MSPC. Note that the uncertainties represented by these methods are different, and the results are therefore not directly interchangeable: one method is not a substitute for the other. See Table 1 for the type of uncertainty represented.

of methods **M4**, **M5**, and **M7**, which consistently converge to values around 0.50 and therefore fail to distinguish between healthy and faulty operating conditions, the remaining approaches exhibit very similar performance levels. In particular, methods **M1**, **M2**, **M3**, **M6**, **M8**, **M9**, and **M10** achieve mean composite indicators in the range of 0.83–0.84, indicating stable and competitive fault detection performance across the 21 faults.

Interestingly, the average composite indicator obtained through several unsupervised strategies slightly exceeds the mean performance observed for both supervised SE and ARD-SE configurations with GPC (Appendix B). This suggests that carefully designed unsupervised kernel selection criteria can provide highly competitive monitoring performance, even without access to labelled fault data.

Nevertheless, this global trend does not hold uniformly across all faults. Certain faults, such as **F21**, demonstrate improved performance under supervised calibration, indicating that the incorporation of labelled information can still be advantageous for specific fault structures.

The results also show that fault detectability remains fault-dependent. Although several unsupervised tuning strategies achieve similar average composite indicators, their behaviour is not identical across all faults. This supports the use of multiple unsupervised kernel-parameter selection criteria when labelled fault data are unavailable, since different criteria emphasize different structural properties of the healthy-operation data. For example, distance-based rules characterize global or local similarity scales, stability-based criteria favour reproducible K-PCA subspaces or control limits, and MMD-based criteria assess temporal consistency of healthy operation. In the proposed probabilistic framework, these unsupervised methods serve as alternative deterministic prior-mean selection mechanisms before Bayesian posterior propagation. Therefore, the role of the unsupervised stage is not to identify a universally optimal tuning rule, but to provide practical calibration routes whose suitability may depend on the fault type and process structure.

TABLE 5. Composite score $((1 - \text{FAR}) + \text{FDR})/2$ for 10 unsupervised kernel-tuning methods across 21 TEP faults (two decimals). The bottom row reports the mean score across faults for each method.

Fault	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10
F01	0.99	0.99	0.99	0.50	0.50	0.99	0.49	0.99	0.99	0.99
F02	0.98	0.98	0.98	0.50	0.50	0.98	0.49	0.98	0.98	0.98
F03	0.53	0.53	0.53	0.50	0.50	0.53	0.49	0.51	0.51	0.53
F04	0.97	0.97	0.94	0.50	0.50	0.92	0.49	0.99	0.99	0.94
F05	0.66	0.65	0.65	0.50	0.50	0.63	0.49	0.64	0.64	0.64
F06	0.99	0.99	0.99	0.50	0.50	0.99	0.49	0.99	0.99	0.99
F07	0.99	0.99	0.99	0.50	0.50	0.99	0.49	0.99	0.99	0.99
F08	0.98	0.98	0.98	0.50	0.50	0.98	0.49	0.98	0.98	0.98
F09	0.49	0.50	0.49	0.50	0.50	0.49	0.49	0.51	0.51	0.49
F10	0.80	0.80	0.79	0.50	0.50	0.79	0.49	0.81	0.81	0.79
F11	0.84	0.86	0.82	0.50	0.50	0.85	0.49	0.87	0.87	0.83
F12	0.98	0.98	0.98	0.50	0.50	0.98	0.49	0.98	0.98	0.98
F13	0.97	0.97	0.96	0.50	0.50	0.97	0.49	0.97	0.97	0.97
F14	0.99	0.99	0.99	0.50	0.50	0.99	0.49	0.99	0.99	0.99
F15	0.63	0.62	0.61	0.50	0.50	0.62	0.50	0.60	0.60	0.60
F16	0.62	0.62	0.62	0.50	0.50	0.61	0.49	0.66	0.66	0.61
F17	0.96	0.96	0.95	0.50	0.50	0.95	0.49	0.97	0.97	0.95
F18	0.94	0.94	0.94	0.50	0.50	0.94	0.49	0.94	0.94	0.94
F19	0.59	0.58	0.59	0.50	0.50	0.60	0.49	0.71	0.71	0.61
F20	0.85	0.85	0.84	0.50	0.50	0.84	0.49	0.87	0.87	0.84
F21	0.70	0.71	0.71	0.50	0.50	0.70	0.49	0.72	0.72	0.70
Mean	**0.83**	**0.83**	**0.83**	**0.50**	**0.50**	**0.83**	**0.49**	**0.84**	**0.84**	**0.83**

TABLE 6. False alarm rates (FAR) and fault detection rates (FDR) for T^2 and SPE charts (SE kernel), rounded to two decimals. Left: deterministic (prior mean). Right: posterior mean. CI is the composite indicator $\text{CI} = \frac{(1-\text{FAR})+\text{FDR}}{2}$, averaged over T^2 and SPE.

Fault	Deterministic (prior mean)					Posterior mean				
	FAR_{T^2}	FDR_{T^2}	FAR_{SPE}	FDR_{SPE}	CI	FAR_{T^2}	FDR_{T^2}	FAR_{SPE}	FDR_{SPE}	CI
F01	0.03	0.66	0.01	0.66	0.82	0.01	0.99	0.01	1.00	**0.99**
F02	0.02	0.98	0.01	0.98	**0.98**	0.00	0.96	0.01	0.99	**0.98**
F03	0.98	0.99	0.97	1.00	**0.51**	0.01	0.00	0.00	0.00	0.50
F04	0.03	0.16	0.02	0.16	0.57	0.02	0.15	0.00	0.88	**0.75**
F05	0.01	0.25	0.01	0.25	0.62	0.02	0.28	0.00	0.26	**0.63**
F06	0.06	0.97	0.02	0.97	0.97	0.02	0.99	0.00	1.00	**0.99**
F07	0.03	0.35	0.01	0.35	0.83	0.01	0.49	0.01	1.00	**0.87**
F08	0.04	0.89	0.01	0.89	0.93	0.00	0.87	0.00	0.98	**0.96**
F09	0.49	0.42	0.44	0.31	0.45	0.00	0.00	0.00	0.00	**0.50**
F10	0.07	0.40	0.02	0.51	**0.71**	0.01	0.26	0.01	0.57	0.70
F11	0.11	0.36	0.06	0.34	0.63	0.01	0.09	0.00	0.60	**0.67**
F12	0.04	0.90	0.02	0.90	0.94	0.01	0.87	0.00	0.99	**0.96**
F13	0.01	0.89	0.00	0.89	0.95	0.00	0.92	0.01	0.96	**0.97**
F14	0.03	0.09	0.01	0.09	0.53	0.01	0.69	0.00	1.00	**0.92**
F15	1.00	1.00	0.95	1.00	0.51	0.00	0.03	0.00	0.10	**0.53**
F16	0.44	0.57	0.45	0.55	**0.56**	0.00	0.04	0.00	0.12	0.54
F17	0.05	0.04	0.03	0.43	0.60	0.00	0.66	0.00	0.90	**0.89**
F18	0.05	0.86	0.02	0.86	0.91	0.00	0.87	0.00	0.90	**0.94**
F19	0.11	0.19	0.06	0.18	**0.55**	0.00	0.01	0.00	0.08	0.52
F20	0.07	0.05	0.01	0.46	0.61	0.00	0.24	0.00	0.59	**0.71**
F21	0.99	1.00	0.97	0.97	0.50	0.00	0.22	0.00	0.40	**0.81**
Mean	**0.22**	**0.57**	**0.19**	**0.56**	**0.71**	**0.01**	**0.46**	**0.00**	**0.68**	**0.79**

The prior mean used for Bayesian sampling was obtained through the unsupervised **M8** method (MMD halves), which selects the kernel lengthscale by minimising the discrepancy between temporal partitions of healthy operation data. As illustrated in Figure 8, the uncertainty bands obtained under the unsupervised calibration are generally wider than those observed in the supervised setting. This reflects the additional epistemic uncertainty introduced when the kernel parameters are inferred without labelled fault information.

Nevertheless, despite the increased posterior spread, the fault remains consistently detected across the entire distribution of posterior draws. In other words, the alarm condition is robust with respect to kernel parameter uncertainty,

TABLE 7. False alarm rates (FAR) and fault detection rates (FDR) for T^2 and SPE charts using an ARD-SE kernel, rounded to two decimals. Left: deterministic (prior mean). Right: posterior mean. Composite indicator (CI) is computed as $\mathrm{CI} = \frac{\mathrm{CI}_{T^2}+\mathrm{CI}_{\mathrm{SPE}}}{2}$ with $\mathrm{CI}_\bullet = \frac{(1-\mathrm{FAR}_\bullet)+\mathrm{FDR}_\bullet}{2}$.

Fault	Deterministic (prior mean)					Posterior mean				
	FAR_{T^2}	FDR_{T^2}	$\mathrm{FAR}_{\mathrm{SPE}}$	$\mathrm{FDR}_{\mathrm{SPE}}$	CI	FAR_{T^2}	FDR_{T^2}	$\mathrm{FAR}_{\mathrm{SPE}}$	$\mathrm{FDR}_{\mathrm{SPE}}$	CI
F01	0.02	0.99	0.00	1.00	0.99	0.01	0.99	0.00	1.00	0.99
F02	0.02	0.98	0.00	0.99	0.99	0.01	0.98	0.00	0.99	0.99
F03	0.01	0.01	0.00	0.01	0.50	0.01	0.01	0.00	0.01	0.50
F04	0.02	0.16	0.01	0.90	0.76	0.02	0.16	0.01	0.90	0.76
F05	0.02	0.25	0.00	0.27	0.63	0.02	0.24	0.00	0.30	0.63
F06	0.01	0.99	0.00	1.00	1.00	0.00	0.99	0.00	1.00	1.00
F07	0.01	0.57	0.01	1.00	0.89	0.01	0.60	0.00	1.00	**0.90**
F08	0.01	0.96	0.00	0.98	0.98	0.01	0.95	0.00	0.98	0.98
F09	0.00	0.00	0.01	0.01	0.50	0.00	0.00	0.01	0.01	0.50
F10	0.01	0.50	0.01	0.47	**0.74**	0.02	0.48	0.00	0.44	0.73
F11	0.01	0.31	0.00	0.61	**0.73**	0.01	0.31	0.00	0.59	0.72
F12	0.01	0.96	0.00	0.99	0.98	0.01	0.96	0.00	0.99	0.98
F13	0.01	0.94	0.01	0.96	0.97	0.01	0.94	0.01	0.96	0.97
F14	0.01	0.82	0.01	1.00	0.95	0.00	0.83	0.01	1.00	**0.96**
F15	0.01	0.20	0.00	0.03	0.55	0.01	0.21	0.00	0.03	**0.56**
F16	0.00	0.10	0.00	0.13	**0.56**	0.00	0.10	0.00	0.12	0.55
F17	0.01	0.77	0.00	0.90	**0.92**	0.01	0.77	0.00	0.90	0.91
F18	0.01	0.89	0.01	0.90	0.94	0.01	0.89	0.01	0.90	0.94
F19	0.00	0.03	0.00	0.12	0.54	0.00	0.03	0.00	0.11	0.54
F20	0.01	0.50	0.00	0.52	0.75	0.01	0.50	0.00	0.52	0.75
F21	0.01	0.30	0.00	0.40	0.67	0.03	0.31	0.00	0.39	0.67
Mean	**0.01**	**0.58**	**0.00**	**0.62**	**0.79**	**0.01**	**0.58**	**0.00**	**0.62**	**0.79**

TABLE 8. Comparison of deterministic and uncertainty-aware K-MSPC approaches for a representative Tennessee Eastman Process fault scenario.

Method	CI	FAR	FDR	Mean T^2 width	Mean SPE width
Deterministic prior-mean K-MSPC	0.98875	0.01875	0.99625	–	–
Bootstrap K-MSPC	0.98562	0.02500	0.99625	0.099285	109.35
Ensemble kernel K-MSPC	0.98875	0.01875	0.99625	0.061735	1581.3
Proposed MCMC posterior K-MSPC	0.98875	0.01875	0.99625	0.025387	1.6375

indicating stable fault discrimination even under unsupervised calibration.

IV. SUMMARY AND FUTURE WORK

This work proposed a probabilistic extension of K-MSPC, with the goal of explicitly quantifying and propagating uncertainty in kernel parameters to the monitoring statistics. The framework follows a two-stage workflow: (i) deterministic kernel calibration to obtain informative prior means and (ii) Bayesian inference of kernel parameters using MCMC, yielding probabilistic control charts for both T^2 and SPE and their corresponding contribution plots.

In the supervised calibration setting, deterministic optimisation of an SE-kernel GPC model produced high classification performance across several optimisers. However, the study highlighted that strong convergence in a supervised classification objective does not necessarily translate into optimal kernel settings for K-PCA monitoring. Nevertheless, due to its favourable accuracy–runtime trade-off, L-BFGS was selected as the preferred deterministic optimiser for the supervised GPC intermediary step.

In the probabilistic stage, uncertainty in kernel parameterswas shown to propagate non-trivially to the control charts. For the SE kernel, posterior-mean control charts generally improved fault detection compared to deterministic prior-mean charts, with notable increases in the composite indicator for several faults (e.g., **F01**, **F14**, **F17**, and **F21**). The uncertainty bands were consistently wider during faulty operating regimes and narrow in the in-control region, indicating that parameter uncertainty is amplified under abnormal process behaviour. The residual-based SPE chart exhibited larger uncertainty than T^2, and the same pattern was reflected in the contribution plots, where variables with higher contributions also displayed wider posterior uncertainty.

When employing an ARD-SE kernel, the uncertainty in both T^2 and SPE charts decreased compared to the isotropic SE case, while preserving the qualitative uncertainty structure (small uncertainty in-control and larger uncertainty during faults). Variable-attribution patterns were largely preserved for T^2, while modest changes in contribution magnitudes and top-ranking variables were observed, particularly in the SPE contributions. In contrast to the SE kernel case, the deterministic and probabilistic ARD-SE results were very similar, suggesting that the additional flexibility of ARD already absorbs a portion of the modelling uncertainty in the deterministic stage.

Finally, an unsupervised calibration path was evaluated using a suite of deterministic kernel selection criteria. Several unsupervised methods achieved competitive average

composite indicators across the benchmark faults. When the **M8** strategy (MMD halves) was used to define the prior mean in the probabilistic workflow, the resulting posterior control charts exhibited wider uncertainty than in the supervised setting, but still achieved robust fault detection across the full set of posterior draws. This demonstrates that probabilistic monitoring is feasible even in the absence of labelled fault data, while providing an explicit quantification of epistemic uncertainty.

The number of retained K-PCA components may also affect posterior uncertainty propagation. Since T^2 represents variation within the retained principal subspace and SPE represents residual variation outside that subspace, the selected dimensionality controls how process variation is partitioned between the two monitoring statistics. Retaining more components can move variation from the residual space into the score space, potentially reducing SPE uncertainty while making T^2 more sensitive to posterior changes in the K-PCA eigenspace. Conversely, retaining fewer components may yield a more stable low-dimensional score representation but leave more variation in the residual space, which can widen SPE credible bands. This effect may be especially pronounced after fault onset, when abnormal observations are projected onto a posterior-dependent nonlinear feature-space geometry. In the present study, the number of retained components was kept fixed across deterministic and probabilistic comparisons in order to isolate the effect of kernel-parameter uncertainty. A systematic investigation of retained dimensionality selection under posterior propagation is therefore a useful direction for future work.

Several directions are recommended to extend the proposed probabilistic K-MSPC framework:

- **A framework to include dynamic process monitoring capabilities.** This would extend the approach to incorporate process dynamics, for example by probabilistic kernel DPCA or dynamic K-PCA formulations, and evaluate whether uncertainty propagation changes under temporal modelling.
- **Faster posterior approximations.** A possibility is to replace MCMC with scalable Bayesian approximations such as variational inference or Laplace-type approximations, with the aim of enabling near real-time uncertainty-aware monitoring.
- **Broader kernel families and structure.** A gap in the present research is the investigation of alternative kernels beyond SE/ARD-SE (e.g., Matérn, rational quadratic, polynomial, and additive or composite kernels) and quantify how kernel choice affects posterior uncertainty and fault interpretability.
- **Calibration objectives aligned with monitoring.** A possibility is to develop deterministic kernel calibration criteria that directly target monitoring performance (e.g., control-limit stability, fault persistence capture, or separation in monitoring statistics), bridging the gap observed between classification-optimal and monitoring-optimal kernel settings.
- **Industrial case studies.** The TEP being a simulated dataset, the framework can be applied to real-world industrial datasets (e.g., spectroscopy and imaging-based monitoring), where sensor drift, missing data, and nonstationarity are present, and evaluate whether probabilistic charts improve robustness and trust in alarm decisions.

V. CONCLUSION

This paper introduced a probabilistic extension of kernel-based multivariate statistical process control, aimed at making kernel parameter uncertainty explicit and actionable in nonlinear process monitoring. Instead of treating kernel parameters as fixed point estimates, the proposed framework infers their posterior distributions and propagates this epistemic uncertainty into the monitoring statistics.

A two-stage workflow was developed. Deterministic calibration (supervised via Gaussian process classification and K-PCR, or unsupervised via healthy-data criteria) provides informative prior means, after which Markov chain Monte Carlo sampling (AM, DRAM, HMC, NUTS) yields posterior draws of kernel parameters. These draws are propagated through kernel PCA to obtain probabilistic T^2 and SPE control charts and uncertainty-aware contribution plots.

Experiments on the Tennessee Eastman Process benchmark demonstrate that kernel parameter uncertainty can materially affect monitoring behaviour. For the isotropic SE kernel, posterior-mean charts frequently improve fault detection compared to deterministic prior-mean charts, and uncertainty bands are consistently small during healthy operation but widen after fault onset, indicating amplified epistemic uncertainty under abnormal regimes. SPE statistics exhibit larger posterior variability than T^2, and the same trend carries to contribution plots, where highly contributing variables also show wider credible intervals. For the ARD-SE kernel, posterior uncertainty is reduced and probabilistic and deterministic results are largely aligned, suggesting that the additional kernel flexibility absorbs part of the uncertainty already at the deterministic stage. Under unsupervised calibration, posterior uncertainty increases, yet fault detection remains robust across posterior draws.

Overall, the proposed probabilistic K-MSPC framework improves robustness and interpretability by attaching uncertainty to both fault detection and variable attribution, and it provides a principled route to uncertainty-aware monitoring in settings where kernel calibration is inherently uncertain.

APPENDIX A NOTATION AND DETERMINISTIC K-PCA/K-MSPC BACKGROUND

Let $\mathbf{X} \in \mathbb{R}^{n\times p}$ denote the autoscaled training data, let $\mathbf{K}$ denote the kernel matrix, and let $\mathbf{K}_c$ denote its centered version. The centered kernel matrix is eigendecomposed as

$$\mathbf{K}_c\mathbf{U} = \mathbf{U}\boldsymbol{\Lambda},$$

where $\boldsymbol{\Lambda} = \mathrm{diag}(\lambda_1, \ldots, \lambda_r)$ contains the retained eigenvalues and $\mathbf{U}$ the corresponding eigenvectors.

For a new sample $\mathbf{x}$, the centered kernel vector is denoted by $\mathbf{k}_c(\mathbf{x})$, and the associated K-PCA score vector is

$$\mathbf{t}(\mathbf{x}) = \mathbf{k}_c(\mathbf{x})^\top \mathbf{U}\boldsymbol{\Lambda}^{-1/2}.$$

The monitoring statistics are the Hotelling-type statistic

$$T^2(\mathbf{x}) = \sum_{h=1}^{r} \frac{t_h(\mathbf{x})^2}{\lambda_h},$$

and the residual statistic

$$\mathrm{SPE}(\mathbf{x}) = k_c(\mathbf{x}, \mathbf{x}) - \sum_{h=1}^{r} t_h(\mathbf{x})^2.$$

To support diagnosis, variable contribution diagnostics are also computed. The contribution of variable d to the T^2 statistic is given by

$$C_d^{T^2}(\mathbf{x}) = \sum_{h=1}^{r} \frac{t_h(\mathbf{x})}{\lambda_h} \frac{\partial t_h(\mathbf{x})}{\partial x_d},$$

which quantifies how variation in variable d affects the modeled K-PCA subspace.

Similarly, the contribution of variable d to the SPE statistic is defined as

$$C_d^{\mathrm{SPE}}(\mathbf{x}) = \frac{\partial \mathrm{SPE}(\mathbf{x})}{\partial x_d}.$$

In the probabilistic formulation proposed in the present work, the same expressions are evaluated for each posterior draw of the kernel parameters, thereby yielding posterior distributions for both monitoring statistics and variable contributions.

APPENDIX B
DETERMINISTIC VS. PROBABILISTIC RESULTS

See Table 6 and Table 7.

APPENDIX C
K-PCR VISUALISATIONS

A. MCMC RESULTS FOR K-PCR

See Fig. 9.

B. PROBABILISTIC CONTROL CHARTS FOR K-PCR

See Fig. 10.

APPENDIX D
ADDITIONAL COMPARISON WITH ALTERNATIVE UNCERTAINTY-AWARE MONITORING APPROACHES

To further position the proposed probabilistic K-MSPC framework relative to other uncertainty-aware monitoring strategies, an additional comparison was carried out for a representative Tennessee Eastman Process fault scenario. In addition to the deterministic prior-mean K-MSPC baseline, two alternative uncertainty-aware approaches were considered: bootstrap K-MSPC and ensemble kernel K-MSPC.

In the bootstrap K-MSPC approach, the healthy calibration observations were repeatedly resampled with replacement, and the K-PCA monitoring model was reconstructed for each bootstrap sample. This provides an estimate of variability due to finite-sample resampling of the calibration data. In the ensemble kernel K-MSPC approach, multiple monitoring models were constructed by perturbing the deterministic log-kernel parameters around the calibrated prior mean, thereby representing variability across a finite collection of nearby kernel configurations. These approaches were compared with the proposed MCMC posterior K-MSPC, which propagates posterior samples of the calibrated kernel parameters through the full K-PCA monitoring pipeline.

As shown in Table 8, the proposed MCMC posterior K-MSPC achieves the same detection performance as the deterministic prior-mean and ensemble kernel baselines, with CI = 0.98875, FAR = 0.01875, and FDR = 0.99625. The bootstrap K-MSPC baseline also performs well, although its composite indicator is slightly lower due to a somewhat higher false alarm rate. These results indicate that the proposed probabilistic framework preserves the strong detection performance of the deterministic K-MSPC baseline while providing an explicit uncertainty quantification layer.

A clear difference is observed in the width of the uncertainty bands. The proposed MCMC posterior K-MSPC produces substantially narrower uncertainty bands than both the bootstrap and ensemble kernel baselines. This is especially evident for the SPE statistic, where the average band width is 1.6375 for the proposed method, compared with 109.35 for bootstrap K-MSPC and 1581.3 for ensemble kernel K-MSPC. This suggests that the MCMC posterior samples provide a more concentrated and targeted representation of uncertainty around the calibrated kernel parameters, whereas bootstrap resampling and local ensemble perturbations introduce substantially broader variability into the resulting K-PCA monitoring model.

Figures 11a and 11b further illustrate the behaviour of the uncertainty-aware methods. It should be noted that the three approaches are not intended to represent identical uncertainty sources and should therefore not be interpreted as directly interchangeable uncertainty estimates. Bootstrap K-MSPC reflects variability induced by resampling the healthy calibration data, ensemble kernel K-MSPC reflects variability across a finite set of perturbed kernel configurations, and the proposed MCMC posterior K-MSPC reflects posterior uncertainty around the calibrated kernel parameters. As a consequence, the methods may converge to different effective kernel geometries and K-PCA eigenspaces, leading to different numerical ranges, particularly for the SPE statistic.

Despite these differences, a consistent qualitative pattern is observed. For all three uncertainty-aware methods, the uncertainty bands are generally narrower during the normal-operation part of the trajectory and increase after the fault is introduced. This indicates that uncertainty in the monitoring model is amplified under abnormal process behaviour, regardless of whether the uncertainty is induced

through bootstrap resampling, local kernel perturbations, or posterior sampling of the calibrated kernel parameters.

The proposed MCMC posterior K-MSPC produces the narrowest uncertainty bands in this representative case, especially for SPE. This should not be interpreted as a direct ranking of the methods, since the uncertainty sources differ. Rather, it indicates that posterior sampling provides a more concentrated representation of uncertainty around the calibrated kernel parameters. In contrast, the bootstrap and ensemble approaches introduce broader variability through resampled calibration sets or locally perturbed kernel configurations. The comparison therefore supports the interpretation that the proposed framework propagates a specific and targeted form of epistemic uncertainty through the nonlinear K-MSPC monitoring model.

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During the preparation of the work, the authors used **Grammarly Pro** in order to improve the readability and check for grammar mistakes. After using this tool, they reviewed and edited the content as needed and take full responsibility for the content of the published article.

CODE AVAILABILITY

The code is publicly available at https://github.com/sab-in-science/probabilistic-fd.

REFERENCES

[1] J. Panaretos and S. Psarakis, "Statistical process monitoring using multivariate methods," *Qual. Rel. Eng. Int.*, vol. 21, no. 4, pp. 397–415, 2005.

[2] H. Saadatinezhad, A. Ramezani, M. Alizadeh, and E. Hajimalek, "Fault tolerant load frequency sharing of a multi-area power system using model predictive control," *J. Eng.*, vol. 2022, no. 3, pp. 337–347, Mar. 2022, doi: 10.1049/tje2.12117.

[3] M. Alizadeh, A. Ramezani, and H. Saadatinezhad, "Fault tolerant control in an unmanned bicycle robot via sliding mode theory," *IET Cyber-Syst. Robot.*, vol. 4, no. 2, pp. 139–152, 2022, doi: 10.1049/csy2.12032.

[4] M. Alizadeh, M. H. Samaei, M. Vahid Estakhri, H. Momeni, and M. T. Beheshti, "Robust trajectory tracking of a 3-DOF robotic arm using a super-twisting fast finite time non-singular terminal sliding mode control in the presence of perturbations," *Proc. Inst. Mech. Eng., I, J. Syst. Control Eng.*, vol. 238, no. 7, pp. 1285–1295, Aug. 2024, doi: 10.1177/09596518241229741.

[5] T. Kourti, "Process analysis, monitoring and diagnosis, using multivariate projection methods," *Chemometric Intell. Lab. Syst.*, vol. 28, no. 1, pp. 3–21, Apr. 1995.

[6] P. Nomikos and J. F. MacGregor, "Multivariate SPC charts for monitoring batch processes," *Technometrics*, vol. 37, no. 1, pp. 41–59, Feb. 1995.

[7] F. Pozzi, "Data science for operations: Integrating statistical process control within industry 4.0 frameworks," *Systems*, vol. 12, no. 3, p. 100, 2024.

[8] B. Schölkopf, A. Smola, and K.-R. Müller, "Nonlinear component analysis as a kernel eigenvalue problem," *Neural Comput.*, vol. 10, no. 5, pp. 1299–1319, Jul. 1998.

[9] L. Liu, J. Liu, H. Wang, S. Tan, M. Yu, and P. Xu, "A multivariate monitoring method based on kernel principal component analysis and dual control chart," *J. Process Control*, vol. 127, Jul. 2023, Art. no. 102994.

[10] W. J. Lee, G. P. Mendis, M. J. Triebe, and J. W. Sutherland, "Monitoring of a machining process using kernel principal component analysis and kernel density estimation," *J. Intell. Manuf.*, vol. 31, no. 5, pp. 1175–1189, Jun. 2020.

[11] B. Zhou and X. Gu, "Multi-block statistics local kernel principal component analysis algorithm and its application in nonlinear process fault detection," *Neurocomputing*, vol. 376, pp. 222–231, Feb. 2020.

[12] R. Tan, J. R. Ottewill, and N. F. Thornhill, "Monitoring statistics and tuning of kernel principal component analysis with radial basis function kernels," *IEEE Access*, vol. 8, pp. 198328–198342, 2020.

[13] M. A. Alam and K. Fukumizu, "Hyperparameter selection in kernel principal component analysis," *J. Comput. Sci.*, vol. 10, no. 7, pp. 1139–1150, Jul. 2014.

[14] Z.-S. Duma, V. Jorry, T. Sihvonen, S.-P. Reinikainen, and L. Roininen, "Optimising kernel-based multivariate statistical process control," *J. Process Control*, vol. 161, May 2026, Art. no. 103679.

[15] A. K. Pani, "Non-linear process monitoring using kernel principal component analysis: A review of the basic and modified techniques with industrial applications," *Brazilian J. Chem. Eng.*, vol. 39, no. 2, pp. 327–344, Jun. 2022.

[16] D. Kim and I.-B. Lee, "Process monitoring based on probabilistic PCA," *Chemometric Intell. Lab. Syst.*, vol. 67, no. 2, pp. 109–123, Aug. 2003.

[17] T. Chen, J. Zhang, and A. J. Morris, "Probabilistic contribution analysis for statistical process monitoring," *Control Eng. Pract.*, vol. 17, no. 8, pp. 879–890, 2009.

[18] Q. Chen, U. Kruger, M. Meronk, and A. Y. T. Leung, "The application of principal component analysis and kernel density estimation to enhance process monitoring," *Control Eng. Pract.*, vol. 8, no. 5, pp. 531–543, 2000.

[19] J. Zhang, H. Chen, S. Chen, and X. Hong, "An improved mixture of probabilistic PCA for nonlinear data-driven process monitoring," *IEEE Trans. Cybern.*, vol. 49, no. 1, pp. 198–210, Jan. 2019.

[20] N. Li and Y. Yang, "Ensemble kernel principal component analysis for improved nonlinear process monitoring," *Ind. Eng. Chem. Res.*, vol. 54, no. 1, pp. 318–329, Jan. 2015, doi: 10.1021/ie503034j.

[21] P. Cui, C. Zhan, and Y. Yang, "Improved nonlinear process monitoring based on ensemble KPCA with local structure analysis," *Chem. Eng. Res. Design*, vol. 142, pp. 355–368, Feb. 2019, doi: 10.1016/j.cherd.2018.12.028.

[22] X. Zheng, X. Yu, X. Yang, and J. J. Rodriguez-Andina, "Practical finite-time command-filtered adaptive backstepping with its applications to quadrotor hovers," *IEEE Trans. Cybern.*, vol. 54, no. 5, pp. 3017–3029, Oct. 2024, doi: 10.1109/TCYB.2023.3323664.

[23] X. Zheng, H. Wen, X. Yang, X. Yu, and J. J. Rodriguez-Andina, "Adaptive neural zeta-backstepping with predefined damping ratio. Application to DC motors," *IEEE Trans. Cybern.*, vol. 55, no. 4, pp. 1701–1705, Apr. 2025, doi: 10.1109/TCYB.2025.3539544.

[24] F. R. Bach and M. I. Jordan, "Kernel independent component analysis," *J. Mach. Learn. Res.*, vol. 3, pp. 1–48, Jul. 2002.

[25] M. Belkin and P. Niyogi, "Laplacian eigenmaps for dimensionality reduction and data representation," *Neural Comput.*, vol. 15, no. 6, pp. 1373–1396, Jun. 2003.

[26] M. Betancourt, "A conceptual introduction to Hamiltonian Monte Carlo," 2017, *arXiv:1701.02434*.

[27] K. M. Borgwardt, C. S. Ong, B. Schölkopf, S. V. N. Vishwanathan, and A. J. Smola, "Integrating structured biological data by kernel maximum mean discrepancy," in *Int. Conf. Res. Comput. Mol. Biol.*, 2006, pp. 78–92.

[28] L. H. Chiang, E. L. Russell, and R. D. Braatz, "Tennessee eastman process," in *Fault Detection and Diagnosis in Industrial Systems*. Cham, Switzerland: Springer, 2001, pp. 103–112.

[29] C. Cortes, M. Mohri, and A. Rostamizadeh, "Algorithms for learning kernels based on centered alignment," *J. Mach. Learn. Res.*, vol. 13, no. 28, pp. 795–828, 2012.

[30] J. J. Downs and E. F. Vogel, "A plant-wide industrial process control problem," *Comput. Chem. Eng.*, vol. 17, no. 3, pp. 245–255, 1993.

[31] S. Forrest, "Genetic algorithms," *ACM Comput. Surv.*, vol. 28, no. 1, pp. 77–80, 1996.

[32] F. Gao and L. Han, "Implementing the nelder-mead simplex algorithm with adaptive parameters," *Comput. Optim. Appl.*, vol. 51, no. 1, pp. 259–277, Jan. 2012.

[33] A. Gretton, K. M. Borgwardt, M. J. Rasch, B. Schölkopf, and A. Smola, "A kernel two-sample test," *J. Mach. Learn. Res.*, vol. 13, pp. 723–773, Mar. 2012.

[34] H. Haario, M. Laine, A. Mira, and E. Saksman, "DRAM: Efficient adaptive MCMC," *Statist. Comput.*, vol. 16, no. 4, pp. 339–354, Dec. 2006.

[35] H. Haario, E. Saksman, and J. Tamminen, "An adaptive Metropolis algorithm," Bernoulli, Tech. Rep., vol. 7, no. 2, pp. 223–242, 2001.

[36] M. Hein and J.-Y. Audibert, "Intrinsic dimensionality estimation of submanifolds in rd," in *Proc. 22nd Int. Conf. Mach. Learn. (ICML)*, 2005, pp. 289–296.

[37] M. D. Hoffman and A. Gelman, "The no-U-turn sampler: Adaptively setting path lengths in Hamiltonian Monte Carlo," *J. Mach. Learn. Res.*, vol. 15, no. 1, pp. 1593–1623, 2014.
[38] H. Kavianihamedani, J. D. Quinn, and J. D. Smith, "New diagnostic assessment of MCMC algorithm effectiveness, efficiency, reliability, and controllability," *IEEE Access*, vol. 12, pp. 42385–42400, 2024.
[39] N. Meinshausen and P. Bühlmann, "Stability selection," *J. Roy. Stat. Soc. B, Stat. Methodol.*, vol. 72, no. 4, pp. 417–473, 2010.
[40] P. Moritz, R. Nishihara, and M. Jordan, "A linearly-convergent stochastic L-BFGS algorithm," in *Proc. 19th Int. Conf. Artif. Intell. Statist.*, 2016, pp. 249–258.
[41] H. Nickisch and C. E. Rasmussen, "Approximations for binary Gaussian process classification," *J. Mach. Learn. Res.*, vol. 9, no. 10, pp. 2035–2078, 2008.
[42] P. Nomikos and J. F. MacGregor, "Monitoring batch processes using multiway principal component analysis," *AIChE J.*, vol. 41, no. 6, pp. 1411–1426, 1995.
[43] H. Owhadi and G. R. Yoo, "Kernel flows: From learning kernels from data into the abyss," *J. Comput. Phys.*, vol. 389, pp. 22–47, Jul. 2019.
[44] R. Rifkin, G. Yeo, and T. Poggio, "Regularized least squares classification," in *Advances in Learning Theory: Methods, Models and Applications*. Amsterdam, The Netherlands: IOS Press, 2003, pp. 131–154.
[45] O. Roy and M. Vetterli, "The effective rank: A measure of effective dimensionality," *IEEE Transactions on Signal Processing*, vol. 55, no. 2, pp. 451–462, Sep. 2007.
[46] B. Schölkopf, J. C. Platt, J. Shawe-Taylor, A. J. Smola, and R. C. Williamson, "Estimating the support of a high-dimensional distribution," *Neural Comput.*, vol. 13, no. 7, pp. 1443–1471, Jul. 2001.
[47] B. Schölkopf and A. J. Smola, *Learning with Kernels: Support Vector Machines, Regularization, Optimization, and Beyond*, Cambridge, MA, USA: MIT Press, 2002.
[48] J. Shawe-Taylor and N. Cristianini, *Kernel Methods for Pattern Analysis*. Cambridge, U.K.: Cambridge Univ. Press, 2004.
[49] F. Simmini, M. Rampazzo, F. Peterle, G. A. Susto, and A. Beghi, "A self-tuning KPCA-based approach to fault detection in chiller systems," *IEEE Trans. Control Syst. Technol.*, vol. 30, no. 4, pp. 1359–1374, Jul. 2022.
[50] E. Schulz, M. Speekenbrink, and A. Krause, "A tutorial on Gaussian process regression: Modelling, exploring, and exploiting functions," *J. Math. Psychol.*, vol. 85, pp. 1–16, Aug. 2018.
[51] H. Babamoradi, F. van den Berg, and Å. Rinnan, "Bootstrap based confidence limits in principal component analysis—A case study," *Chemometric Intell. Lab. Syst.*, vol. 120, pp. 97–105, Jan. 2013.
[52] D. M. J. Tax and R. P. W. Duin, "Support vector data description," *Mach. Learn.*, vol. 54, no. 1, pp. 45–66, 2004.
[53] L. Zelnik-Manor and P. Perona, "Self-tuning spectral clustering," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 17, 2004, pp. 1601–1608.

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